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FUSED AMINO PYRIMIDINE COMPOUNDS FOR TREATMENT OF NEUROPATHIC AND NEURO-INFLAMMATORY PAIN

Abstract

Compositions and methods for treating neuropathic pain or neuro-inflammatory pain with a compound of Formula (I), or pharmaceutically acceptable salts thereof, are provided. The compositions and methods may be used to improve one or more symptoms of neuropathic pain or neuro-inflammatory pain.
##STR00001##
or pharmaceutically acceptable salts thereof, wherein R.sup.1, R.sup.2, R.sup.7 and ring A have any of the meanings herein defined in the description.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application claims benefit of and priority to the U.S. Provisional Application No. 63/561,915, filed on Mar. 6, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Treatment of neuropathic pain and neuroinflammatory pain with fused amino pyrimidine compounds.

BACKGROUND

[0003] Pain may be defined as a localized or generalized unpleasant bodily sensation or complex of sensations that causes mild to severe discomfort and emotional distress. According to the National Institutes of Health, National Institute of Neurological Disorders and Stroke, pain can be classified as acute or chronic. Several different types of pain make up chronic pain, including inflammatory pain following tissue injury (e.g., sprains, broken bones, bruises, arthritis, etc.), cancer pain, neuropathic pain and neuro-inflammatory pain following nerve injury, spinal cord injury and brain injury (e.g., stroke and trauma).

[0004] Neuropathic pain is defined by the International Association for the Study of Pain (IASP) as pain caused by a lesion or disease of the somatosensory nervous system. Neuropathic pain may be characterized by burning pain, paresthesia (a sensation of tingling, tickling, prickling and pricking) and dysesthesia (an unpleasant, abnormal sense of touch). Chronic neuropathic pain is a consequence of injury or disease affecting the somatosensory system that persists for more than three months, often after the cause is no longer active. According to the IASP, at least 7% to 10% of the general population suffers from neuropathic pain, although prevalence may be underestimated. Treatment remains suboptimal because traditional drugs including opiates provide only modest and often tachyphylactic pain relief.

[0005] Neuroinflammation is typically a localized form of inflammation that occurs in the peripheral nervous system (PNS, including peripheral nerves and ganglia) and central nervous system (CNS, including the spinal cord and brain). Ji, R R et al., Neuroinflammation and Central Sensitization in Chronic and Widespread Pain. *Anesthesiology.* 2018 August; 129(2):343-366. Characteristic features of neuroinflammation are 1) vasculature changes that result in increased vascular permeability, 2) infiltration of leukocytes, 3) activation of glial cells, and 4) production of inflammatory mediators including cytokines and chemokines. Id. Sustained increase of cytokines and chemokines in the CNS promotes chronic widespread neuro-inflammatory pain that affects multiple body sites. Id. Thus, neuroinflammation drives widespread neuro-inflammatory chronic pain via central sensitization. Id.

[0006] Spinal cord injury, peripheral nerve injury and traumatic brain injury, e.g., stroke, may cause neuropathic pain and/or neuroinflammatory pain. Diseases such as multiple sclerosis, amyotrophic lateral sclerosis, Huntington's disease, sickle cell anemia and diabetes may also cause neuropathic pain and/or neuroinflammatory pain.

[0007] In many instances pharmacologic treatments for pain may offer some relief but are far from a panacea. The hazards of opiate treatment are well-known. Addiction resulting from chronic administration of opiates has become a national and global scourge. Chronic corticosteroid administration may cause increased appetite and weight gain, changes in mood, muscle weakness and Cushing syndrome. Non-steroidal anti-inflammatory drugs are associated with side-effects such as stomach ulcers and tinnitus. Analgesics such as acetaminophen may cause liver damage. There remains a need for additional pharmaceutical therapies to treat pain.

[0008] Fused amino pyridine compounds and pharmaceutically acceptable salts thereof are disclosed in WO2021/180952, incorporated herein by reference. As described in WO2021/180952, these compounds and their pharmaceutically acceptable salts selectively modulate KCC2 and are used to treat or prevent KCC2 mediated disease, including neurological disorders. As described in the OMIM database, the LC12A5 gene encodes the neuronal KCC2 channel that is the major extruder of intracellular chloride in mature neurons. In the presence of low intraneuronal chloride, the binding of GABA and glycine to their ionotropic receptors results in chloride influx with subsequent hyperpolarization contributing to neuronal inhibition.

[0009] Hyperpolarizing GABA.sub.AR currents are critically dependent upon efficient Cl.sup.− extrusion, which is facilitated by KCC2. Conditional inactivation of KCC2 in the hippocampus leads to neuronal Cl.sup.− accumulation and depolarizing GABA.sub.AR currents. These deficits in inhibition parallel the development of spontaneous seizures, neuronal apoptosis and reactive astrocytosis (Kelley et al., EBioMedicine 32, 62-71 (2018)). GABA.sub.A signaling is a major inhibitory neurotransmitter mechanism in the adult brain and consequently KCC2 has a key role in normal neurodevelopment and various neurological disorders. Decreased activity of KCC2 has been implicated in the pathogenesis of neurological disorders including epilepsy (Galanopoulou et al, Epilepsia 2007; 48:14-18; Huberfield et al, The Journal of Neuroscience (2007) 27, 9866-9873), neuropathic pain (Price et al, Curr Top Med Chem 2005; 5:547-555), Rett's syndrome (Tang et al, 2019, Translational Medicine, 11(503)), autism (Tyzio et al, Science 343, 675-679, Merner et al, Frontiers in cellular neuroscience 9, 2015), mental disorders, spinal cord injury (Boulenguez et al, Nature Medicine 2010, 16, 302-307) and conditions in which there is neuronal hyperexcitability such as ALS (Fuchs et al, Journal of Neuropathology & Experimental Neurology, Volume 69, Issue 10, October 2010, Pages 1057-1070).

[0010] Direct modulation of KCC2 by interaction with small molecules has been reported. Delpire et al (Proc Natl Acad Sci USA. 2009 Mar. 31; 106(13): 5383-5388) describe an assay to identify small molecule inhibitors of KCC2 and Zhang et al (Journal of Biomolecular Screening 15(2): 2010) describe an assay used to identify positive modulators of KCC2.

SUMMARY

[0011] Methods and compositions for treating neuropathic pain or neuro-inflammatory pain are provided and, in embodiments, include administering to a subject diagnosed with neuropathic pain or neuro-inflammatory pain an effective amount of Formula (I):

##STR00002## [0012] or a pharmaceutically acceptable salt thereof, wherein: [0013] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; [0014] R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0015] A is selected from

##STR00003## [0016] or a N-oxide thereof; [0017] R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; [0018] R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0019] R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; [0020] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0021] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0022] R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; [0023] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl; [0024] R.sup.10 is —C.sub.1-6alkyl; [0025] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0026] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0027] m is 0 or 1; and [0028] n is 1, 2 or 3.

[0029] In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement in two or more symptoms of the neuropathic pain or neuro-inflammatory pain. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement in next day functioning of the subject. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating neuropathic pain or neuro-inflammatory pain in a subject. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in providing improvement in next day functioning of a subject with neuropathic pain or neuro-inflammatory pain. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for the manufacture of a medicament for treatment of neuropathic pain or neuro-inflammatory pain.

[0030] In embodiments, a compound of Formula (I) is Compound A:

##STR00004##

[0031] In embodiments, a compound of Formula (I) is Compound B:

##STR00005##

[0032] In embodiments, a compound of Formula (I) is Compound C:

##STR00006##

[0033] In embodiments, a compound of Formula (I) is Compound D:

##STR00007##

[0034] In embodiments, a compound of Formula (I) is Compound E:

##STR00008##
[0035] In embodiments, a compound of Formula (I) is Compound F:
##STR00009##
[0036] In embodiments, a compound of Formula (I) is Compound G:
##STR00010##
[0037] In embodiments, a compound of Formula (I) is Compound H:
##STR00011##

Description

BRIEF DESCRIPTION OF THE FIGURES
[0038] FIG. **1**A is a drug screening workflow chart illustrating the steps taken to identify KCC2 direct activators, a multi-tiered program starting from a compound library of 1.3 million compounds. Identification of a subseries of fused pyrimidine ring compounds, denoted as subseries A (SSA) were identified as lead-hits from the compound library screen.
[0039] FIG. **1**B depicts immunoblotting results for HEK-293 cells expressing KCC2 exposed to vehicle (−) or 30 μM SSA1 (+) for 90 min and then heated to 37-58° C. Soluble fractions were subsequently immunoblotted with KCC2 and GAPDH antibodies.
[0040] FIG. **1**C is a graph depicting thermal aggregation curves for KCC2 under control conditions and in the presence of SSA-0126. *=significantly different to control p<0.001; t-test; n=4 transfections. The level of remaining soluble native KCC2 at each temperature was normalized to that seen at 37° C. (I×I37° C.).
[0041] FIG. **1**D is a graph depicting isothermal dose response curves constructed for destabilization of KCC2 by SSA1 at 50° C., n=4 transfections
[0042] FIG. **1**E shows the structure of Compound A (also referred to as CmpA).
[0043] FIG. **1**F is a graph showing the effects of Compound A on KCC2 activity expressed in HEK-293 cells as measured using TI flux. Data is normalized to values seen with vehicle (100%), n=3.
[0044] FIG. **1**G is a chart showing the EC.sub.50 of Compound A for KCC2, KCC2-S940A, KCC3, KCC4 and NKCC1 as measured in expressing HEK-293 cells, n=4 transfections. In all panels p values were determined using unpaired t-tests with Welch's correction.
[0045] FIG. **2**A depicts three graphs showing E.sub.Gly values versus time as measured from HEK-293 cells expressing KCC2 together with GlyRα1 using perforated patch clamp recordings. Individual shifts in E.sub.Gly are shown for cells incubated with Compound A (0.3 and 3 μM) or vehicle (V) for 15 min.
[0046] FIG. **2**B depicts a first bar graph showing ΔE.sub.Gly (mM) by concentration of vehicle and Compound A and a second bar graph showing reduction in [Cl.sup.−](mM) by concentration of vehicle and Compound A. Mean shifts in E.sub.Gly were determined for cells treated with vehicle, p=0.1227, Compound A 0.3 μM (*p=0.0051), and 3 μM (*p=0.0350), respectively, n=5 transfections. Compound A induced changes in [Cl.sup.−] were calculated from E.sub.Gly using the Nernst equation for vehicle (V), (p=0.1227), 300 nM, (*p=0.0051), or 3 μM (*p=0.0350) CmpA, n=5 transfections.
[0047] FIG. **2**C depicts immunoblotting results and a bar graph showing surface/total KCC2 (% conc.) by concentration of Compound A. HEK-293 cells (UT) or those expressing KCC2 were exposed to V, 0.3 μM, or 3 μM 350 for 15 min and biotinylated with NHS-biotin. Surface and total extracts were subsequently immunoblotted with KCC2 and actin antibodies. The ratio of surface/total KCC2 immunoreactivity was determined and normalized to V (100%) (n=3 transfections).
[0048] FIG. **2**D depicts immunoblotting results obtained by exposing untransfected HEK-293 cells

(U) or those expressing KCC2 to vehicle (V), 0.3, or 3 μM, Compound A, for 15 min and biotinylated with NHS-Biotin. Surface and total extracts were subsequently immunoblotted with KCC2 and actin antibodies. The ratio of surface/total KCC2 immunoreactivity was determined and normalized to V (100%) for 0.3 (*p=0.659) or 3 μM Compound A (*p=0.739), n=3 transfections.

[0049] FIG. **2**E depicts a first bar graph showing pS940/KCC2 (% concentration) of Compound A (μM) and a second bar graph showing pT1007/KCC2 (% concentration) of Compound A (μM). HEK-293 cells were treated with V (−) or 3 μM Compound A (+) for 15 min. Cell lysates were immunoblotted with KCC2, pS940, KCC2 and actin antibodies. The ratios of pS940/KCC2 (p=0.491) and pT1007/KCC2 (p=0.721) were then compared to V (100%), n=3 transfections. In all panels p values were determined using unpaired t-tests with Welch's correction.

[0050] FIG. **3**A depicts a graph showing current (pA) versus holding potential (mV) for vehicle and Compound A. 18-21 Div hippocampal neurons were subjected to gramicidin perforated patch clamp recordings in the presence of bumetanide (10 μM) and TTX (500 nM). After attaining perforation, cultures were exposed to 300 nM Compound A, or V (1% BCD) for 15 min. E.sub.GABA was then determine using voltage ramps protocols and representative current-voltage (I-V) plots are shown for neurons at 0 (dark gray) and 15 (light gray) min treatment for V and Compound A treated neurons.

[0051] FIG. **3**B depicts a first bar graph showing neuronal E.sub.GABA values by exposure to vehicle (V) over time and a second bar graph defining neuronal E.sub.GABA values by exposure to Compound A over time. E.sub.GABA values were measured at 0 and 15 min following treatment with V (p=0.2190) or 300 nM Compound A (*p=0.0104), n=5 cultures.

[0052] FIG. **3**C depicts a bar graph showing neuronal [Cl.sup.−] values by exposure to Compound A over time. [Cl.sup.−] values were calculated from E.sub.GABA values for V (p=0.1646) or Compound A (*p=0.0153) treated neurons, n=5 cultures.

[0053] FIG. **3**D depicts a first bar graph showing neuronal Basal E.sub.GABA (mV) upon exposure to vehicle (V) and Compound A, and a second bar graph showing neuronal 11K A E.sub.GABA upon exposure to vehicle (V) and Compound A. 18-21 Div hippocampal neurons were incubated with 300 nM Compound A or vehicle for 1 h. Neurons were subject to whole cell recording using an intracellular solution containing 30 mM Cl.sup.−. 5 min later basal E.sub.GABA values were determined and compared between treatments, *p=0.0253, n=4 cultures. Neurons were subsequently exposed to 10 μM VU0463271 and the magnitude of the shift in E.sub.GABA (VU0463271ΔE.sub.GABA) was then determined and compared between groups, *p=0.040 n=4 cultures.

[0054] FIG. **4**A depicts a bar graph showing accumulation of Compound A in the mouse brain within 30 minutes after intravenous injection of 25 mg/kg Compound A in 5% β-cyclodextrin (BCD). 15 and 60 minutes later drug levels were measured via LC-MS/MS, n=3 mice.

[0055] FIG. **4**B depicts a bar graph showing accumulation of Compound A in the mouse brain after intravenous injection of 50 mg/kg Compound A in 5% β-cyclodextrin (BCD). 1 and 2 h later drug levels were measured via LC-MS/MS, n=3 mice.

[0056] FIG. **4**C is a graph showing accumulation of Compound A in the mouse brain within 30 minutes after subcutaneous injection of 50 mg/Kg Compound A in 5% β-cyclodextrin (BCD). Drug accumulation was then measured over a time course of 8 h via LC-MS/MS, n=3-4 mice.

[0057] FIG. **4**D depicts a first bar graph showing the effect of Compound A and vehicle (V) on distance traveled after mice were dosed with vehicle (V) or 50 mg/kg Compound A and a second bar graph showing the effect of Compound A and vehicle (V) on time spent in a center zone after mice were dosed with vehicle or 50 mg/kg Compound A. 2-3 h after injection, the mice were placed in the center of a 60 cm×60 cm open field and allowed to explore for 10 min. The total distance traveled (p=0.365) and time in the center of the area (p=0.425) were then quantified and compared between treatments using t-tests; n=9 mice.

DETAILED DESCRIPTION

[0058] Compositions and methods of treating neuropathic pain or neuro-inflammatory pain are provided and, in embodiments, include administering to a subject in need thereof an effective amount Formula (I):
##STR00012##
or a pharmaceutically acceptable salt thereof, wherein: [0059] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; [0060] R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0061] A is selected from ##STR00013## [0062] or a N-oxide thereof; [0063] R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; [0064] R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0065] R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; [0066] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; [0067] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0068] R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; [0069] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl; [0070] R.sup.10 is —C.sub.1-6alkyl; [0071] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0072] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0073] m is 0 or 1; and [0074] n is 1, 2 or 3.
[0075] In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement

in one or more symptoms of the neuropathic pain or neuro-inflammatory pain. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement in two or more symptoms of the neuropathic pain or neuro-inflammatory pain. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain include administering a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, to a subject diagnosed with neuropathic pain or neuro-inflammatory pain to provide improvement in next day functioning of the subject. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in treating neuropathic pain or neuro-inflammatory pain in a subject. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in providing improvement in next day functioning of a subject with neuropathic pain or neuro-inflammatory pain. In embodiments, provided herein is a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for the manufacture of a medicament for treatment of neuropathic pain or neuro-inflammatory pain. In embodiments, the pain being treated is neuropathic pain. In embodiments, the pain being treated is neuro-inflammatory pain.

[0076] Without wishing to be bound by any particular theory, neuropathic pain or neuro-inflammatory pain can result from impaired Cl.sup.− transport. See, e.g., Li et al., 2016, Cell Reports 15, 1376-1383. KCC2 is a K.sup.+—Cl.sup.− cotransporter and responsible for maintaining low Cl.sup.− concentration in neurons of the central nervous system (CNS), essential for postsynaptic inhibition through GABA.sub.A and glycine receptors. As a result of their KCC2 activation activity, the compounds of Formula (I), or pharmaceutically acceptable salts thereof, restore GABAergic function in patients experiencing neuropathic pain or neuro-inflammatory pain involving impaired Cl.sup.− transport, thereby relieving, alleviating or eliminating symptoms.

[0077] Symptoms of neuropathic pain or neuro-inflammatory pain can include localized or generalized unpleasant bodily sensations, allodynia, hyperalgesia, burning, paresthesia and dysesthesia. In embodiments, the pain is chronic pain. In embodiments, the pain is acute pain.

[0078] The compounds of Formula (I), and pharmaceutically acceptable salts thereof, provide prolonged and effective pain relief without the addiction potential of opiates or the side effects associated with non-steroidal anti-inflammatory medications (NSAIDs) or the side effects associated with corticosteroids. Accordingly, patients who are prescribed opiates, corticosteroids or NSAIDs can avoid, or receive lower doses of these medications when they are co-administered with the compounds of Formula (I), and pharmaceutically acceptable salts thereof. For patients who have become addicted to opiates, the compounds of Formula (I), and pharmaceutically acceptable salts thereof provide an effective modality for weaning off the opiates.

[0079] In embodiments, the terms “effective amount” or “therapeutically effective amount” as applied to a treating neuropathic pain or neuro-inflammatory pain refer to an amount of a compound, material, composition, medicament, or other material that is effective to achieve a particular pharmacological and/or physiologic effect in connection with symptoms of neuropathic pain or neuro-inflammatory pain described above.

[0080] In embodiments, effective amount refers to an amount which may be suitable to prevent a decline in or exacerbation of any one or more of the above symptoms, or, in embodiments, to improve any one or more of the above symptoms. In embodiments, effective amount refers to an amount which may be suitable to prevent a decline in two or more of the above symptoms, or, in embodiments, to improve two or more of the above symptoms. In embodiments, effective amount refers to an amount which may be suitable to prevent a decline in three or more of the above symptoms, or, in embodiments, to improve three or more of the above symptoms. In embodiments, an effective amount may be suitable to reduce either the extent or rate of decline in a subject's cognitive skills or functioning, and/or the effective amount may be suitable to delay the onset of such decline. Such effectiveness may be achieved, for example, by administering compositions described herein to an individual or to a population. In embodiments, the reduction, or delay of such a decline, or the improvement in an individual or population can be relative to a cohort, e.g., a control subject or a cohort population that has not received the treatment or been administered the composition or medicament. The terms “effective amount” and “therapeutically effective amount” are used interchangeably herein.

[0081] The dosage amount can vary according to a variety of factors such as subject-dependent variables (e.g., age, immune system, health, etc.), the disease or disorder being treated, as well as the route of administration and the pharmacokinetics of the agent being administered.

[0082] Many pharmaceutical products are administered as a fixed dose, at regular intervals, to achieve therapeutic efficacy. Duration of action is typically reflected by a drug's plasma half-life. Since efficacy is often dependent on sufficient exposure within the central nervous system administration of CNS drugs with a short half-life may require frequent maintenance dosing. CNS drugs with a long half-life may require less frequent maintenance dosing.

[0083] It should be understood that in the context of Formula (I) and other Formulas disclosed herein that unless otherwise indicated, the term “alkyl” includes both linear and branched chain alkyl groups. The prefix C.sub.p-q in C.sub.p-q alkyl and other terms (where p and q are integers) indicates the range of carbon atoms that are present in the group, for example C.sub.1-3alkyl includes C.sub.1alkyl (methyl), C.sub.2alkyl (ethyl) and C.sub.3alkyl (propyl as n-propyl and isopropyl).

[0084] The term “C.sub.p-q alkoxy” comprises —O—C.sub.p-q alkyl groups and —C.sub.p-q alkyl groups where the O atom is within the alkyl chain, for example, —CH.sub.2—O—CH.sub.3.

[0085] The term “C.sub.p-q alkenyl” includes both linear and branched chain alkyl groups containing at least two carbon atoms and at least one double carbon-carbon bond.

[0086] The term “C.sub.p-q alkenyloxy” comprises —O—C.sub.p-q alkenyl groups and —C.sub.p-q alkenyl groups where the O atom is within the alkenyl chain.

[0087] The term “C.sub.p-q alkynyl” includes both linear and branched chain alkyl groups containing at least two carbon atoms and at least one triple carbon-carbon bond.

[0088] The term “C.sub.p-q alkynyloxy” comprises —O—C.sub.p-q alkynyl groups and —C.sub.p-q alkynyl groups where the O atoms is within the alkynyl chain.

[0089] C.sub.p-q cycloalkyl refers to a cyclic non-aromatic group of p-q carbon atoms and no heteroatoms. For example, a 3 to 7 membered cycloalkyl refers to a ring containing 3 to 7 carbon atoms. Suitable C.sub.3-7cycloalkyl rings include cyclopropyl, cyclobutyl, cyclopentyl and cyclohexyl.

[0090] Aryl is a 6 to 10 membered monocyclic or bicyclic aromatic ring containing no heteroatoms. Aryl includes phenyl.

[0091] Heterocycloalkyl is a monocyclic saturated or partially unsaturated, non-aromatic ring having, for example, 3 to 7 members, such as 3 to 6 members, 5 to 7 members such as 5 or 6 members, where at least one member and up to 4 members, particularly 1, 2 or 3 members of the ring are heteroatoms selected from N, O and S, and the remaining ring atoms are carbon atoms, in stable combinations known to those of skill in the art. Heterocycloalkyl ring nitrogen and sulphur atoms are optionally oxidised. Suitable heterocycloalkyl rings include morpholinyl, thiazolidinyl, homomorpholine, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and tetrahydrofuranyl. In embodiments, when R.sup.7 is heterocycloalkyl, optionally two substituents on the same ring carbon together with the carbon to which they are attached form a 5 to 7 membered heterocycloalkyl ring, thereby creating a spirocyclic ring system. For example, in embodiments, R.sup.7 is morpholinyl and two substituents on the same ring carbon together form a tetrahydropyran.

[0092] Heteroaryl is a polyunsaturated, monocyclic 5 or 6 membered aromatic ring containing at least one and up to 3 heteroatoms, particularly, 1 or 2 heteroatoms selected from N, O and S, and the remaining ring atoms are carbon atoms. Heteroaryl ring nitrogen and sulphur atoms are optionally oxidised. Suitable heteroaryl rings include pyridinyl, isoxazolyl, oxadiazolyl, imidazolyl, pyrazinyl, oxazolyl, thiophenyl and thiazolyl.

[0093] The term “halo” is fluorine, chlorine or bromine.

[0094] The use of the dashed bond “custom-character” in rings A of Formula (I) represents the fusion of the pyrimidine ring.

[0095] Where the term “optionally” is used, it is intended that the subsequent feature may or may not occur. As such, use of the term “optionally” includes instances where the feature is present, and also instances where the feature is not present. For example, a group “optionally substituted with 1, 2 or 3 -F substituents” includes group with and without an —F substituent.

[0096] The term “substituted” means that one or more hydrogens (for example 1 or 2 hydrogens, or alternatively 1 hydrogen) on the designated group is replaced by the indicated substituent(s) (for example 1, 2 or 3 substituents, or alternatively 1 or 2 substituents, or alternatively 1 substituent), provided that any atom(s) bearing a substituent maintains a permitted valency. Substituent combinations encompass only stable compounds and stable synthetic intermediates. “Stable” means that the relevant compound or intermediate is sufficiently robust to be isolated and have utility either as a synthetic intermediate or as an agent having potential therapeutic utility. If a group is not described as “substituted”, or “optionally substituted”, it is to be regarded as unsubstituted (i.e. that none of the hydrogens on the designated group have been replaced).

[0097] The term “pharmaceutically acceptable” is used to specify that an object (for example a salt, dosage form or excipient) is suitable for use in patients. An example list of pharmaceutically acceptable salts can be found in the *Handbook of Pharmaceutical Salts: Properties, Selection and Use*, P. H. Stahl and C. G. Wermuth, editors, Weinheim/Ztirich:Wiley-VCH/VHCA, 2002. In embodiments, a suitable pharmaceutically acceptable salt of a compound of the Formula (I) is, for example, a salt formed within the human or animal body after administration of a compound of the Formula (I), to said human or animal body.

[0098] In embodiments, a suitable pharmaceutically acceptable salt of a compound of Formula (I) is, for example, an acid addition salt. An acid addition salt of a compound of Formula (I) may be formed by bringing the compound into contact with a suitable inorganic or organic acid under conditions known to the skilled person. In embodiments, compounds described herein may form base addition salts. A base-addition salt of a compound of Formula (I) may be formed by bringing the compound into contact with a suitable inorganic or organic base under conditions known to the skilled person.

[0099] In embodiments, Formula (I) may be provided as an acid addition salt, a zwitter ion hydrate, zwitter ion anhydrate, hydrochloride or hydrobromide salt, or in the form of the zwitter ion monohydrate. Acid addition salts, include but are not limited to, maleic, fumaric, benzoic, ascorbic, succinic, oxalic, bis-methylenesalicylic, methanesulfonic, ethane-disulfonic, acetic, propionic, tartaric, salicylic, citric, gluconic, lactic, malic, mandelic, cinnamic, citraconic, aspartic, stearic, palmitic, itaconic, glycolic, pantothenic, p-amino-benzoic, glutamic, benzene sulfonic or theophylline acetic acid addition salts, as well as the 8-halotheophyllines, for example 8-bromo-theophylline. In embodiments, inorganic acid addition salts, including but not limited to, hydrochloric, hydrobromic, hydroiodic, sulfuric, sulfamic, phosphoric or nitric acid addition salts may be used.

[0100] In embodiments, there is provided a compound of Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, there is provided a compound of Formula (I). In embodiments there is provided a pharmaceutically acceptable salt of a compound of Formula (I).

[0101] Compounds and salts described herein may exist in solvated forms and unsolvated forms. For example, a solvated form may be a hydrated form, such as a hemi-hydrate, a monohydrate, a dihydrate, a trihydrate or an alternative quantity thereof. The description herein encompasses all such solvated and unsolvated forms of compounds of Formula (I), particularly to the extent that such forms possess KCC2 modulating activity, as for example measured using the tests described herein.

[0102] The following embodiments of moiety A may be applied to the description of the compounds of Formula (I), provided herein:

[0103] Moiety A is selected from:

##STR00014##

or a N-oxide thereof.

[0104] In embodiments, moiety A is

##STR00015##

[0105] In embodiments, moiety A is

##STR00016##

[0106] In embodiments, moiety A is

##STR00017##

[0107] In embodiments, moiety A is

##STR00018##

[0108] In embodiments, moiety A is

##STR00019##

[0109] In embodiments, there is provided a compound of Formula (II):

##STR00020##

or a pharmaceutically acceptable salt thereof, wherein R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b and R.sup.7 are as defined for Formula (I).

[0110] In embodiments, there is provided a compound of Formula (II) or a pharmaceutically acceptable salt thereof, wherein R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b and R.sup.7 are as defined for Formula (I) and when R.sup.7 is morpholinyl, either: [0111] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; heteroaryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and aryl is substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; or [0112] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; and heteroaryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy, wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and R.sup.2 is selected from -halo and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3.

[0113] In embodiments, there is provided a compound of Formula (II) or a pharmaceutically acceptable salt thereof, wherein: [0114] R.sup.1 is selected from —C.sub.2-6alkyl; —C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; phenyl optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3 alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; —O-phenyl optionally substituted with 1 or 2-halo substituents; —O—CH.sub.2-phenyl; and thiophenyl; wherein —C.sub.2-6alkyl and —C.sub.2-6alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0115] R.sup.2 is selected from —H, —F and —CH.sub.3; [0116] R.sup.3 is selected from —C.sub.2-4alkynyl and —C.sub.1-3alkyl optionally substituted with —NR.sup.8R.sup.9; [0117] R.sup.4a and R.sup.4b are both —H; [0118] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 substituents selected from —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy; cyclopropyl; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl; [0119] R.sup.8 and R.sup.9 are each independently selected from —C.sub.1-6alkyl; [0120] R.sup.10 is selected from —C.sub.1-3alkyl; [0121] R.sup.11 is selected from —C.sub.1-3alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3 alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0122] R.sup.12 is selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0123] n is 1 or 2.

[0124] In embodiments, there is provided a compound of Formula (II) or a pharmaceutically acceptable salt thereof, wherein: [0125] R.sup.1 is selected from —C.sub.2-6alkyl; —C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; phenyl substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; —O-phenyl optionally substituted with 1 or 2-halo substituents; —O—CH.sub.2-phenyl; and thiophenyl; wherein —C.sub.2-6alkyl and —C.sub.2-6alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; and R.sup.2 is selected from —H, —F and —CH.sub.3; or [0126] R.sup.1 is selected from —C.sub.2-6alkyl; —C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; unsubstituted phenyl; —O— phenyl optionally substituted with 1 or 2-halo substituents; —O—CH.sub.2-phenyl; and thiophenyl; wherein —C.sub.2-6alkyl and —C.sub.2-6alkoxy are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; and R.sup.2 is selected from —F and —CH.sub.3; [0127] R.sup.3 is selected from —C.sub.2-4alkynyl and —C.sub.1-3alkyl optionally substituted with —NR.sup.8R.sup.9; [0128] R.sup.4a and R.sup.4b are both —H; [0129] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 substituents selected from —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy; cyclopropyl; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl; [0130] R.sup.8 and R.sup.9 are each independently selected from —C.sub.1-6alkyl; [0131] R.sup.10 is selected from —C.sub.1-3alkyl; [0132] R.sup.11 is selected from —C.sub.1-3alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0133] R.sup.12 is selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; and [0134] n is 1 or 2.

[0135] In embodiments, there is provided a compound of Formula (III):

##STR00021##

or a pharmaceutically acceptable salt thereof, wherein R.sup.1, R.sup.2, R.sup.5a, R.sup.5b, R.sup.5c, R.sup.5d and R.sup.7 are as defined for Formula (I).

[0136] In embodiments, there is provided a compound of Formula (III) or a pharmaceutically acceptable salt thereof, wherein:

[0137] R.sup.1 is selected from C.sub.3-7cycloalkyl and C.sub.6-10aryl, wherein the aryl is optionally substituted with a —C.sub.2-8alkoxy substituent wherein the alkoxy is optionally substituted with 1 or 2 —CF.sub.3 substituents; [0138] R.sup.2 is —H; [0139] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each —H; [0140] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H.

[0141] In embodiments, there is provided a compound of Formula (III) or a pharmaceutically acceptable salt thereof, wherein: [0142] R.sup.1 is selected from C.sub.3-7cycloalkyl and C.sub.6-10aryl, wherein the aryl is optionally substituted with a —C.sub.2-8alkoxy substituent wherein the alkoxy is optionally substituted with 1 or 2 —CF.sub.3 substituents; [0143] R.sup.2 is —H; [0144] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each —H; [0145] R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl and a 5 or 6 membered monocyclic heteroaryl, wherein the heterocycloalkyl and heteroaryl are optionally substituted with a substituent selected from —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —OH; and cyclopropyl.

[0146] In embodiments, there is provided a compound of Formula (IV):

##STR00022##

or a N-oxide or pharmaceutically acceptable salt thereof, wherein R.sup.1, R.sup.2, R.sup.6 and R.sup.7 are as defined for Formula (I).

[0147] In embodiments, there is provided a compound of Formula (IV), or a N-oxide or pharmaceutically acceptable salt thereof, wherein:

[0148] R.sup.1 selected from C.sub.3-7cycloalkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0149] R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0150] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0151] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-

5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; [0152] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl; [0153] R.sup.10 is —C.sub.1-6alkyl; [0154] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; [0155] R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; [0156] n is 1, 2 or 3.

[0157] In embodiments, there is provided a compound of Formula (IV), or a N-oxide or pharmaceutically acceptable salt thereof, wherein:

[0158] R.sup.1 selected from C.sub.3-7cycloalkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; [0159] R.sup.2 is —H; [0160] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl; [0161] R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl optionally substituted with 1, 2 or 3 groups selected from —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; and C.sub.3-5cycloalkyl; [0162] R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6alkyl.

[0163] The following embodiments of moieties R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b, R.sup.4c, R.sup.4d, R.sup.5, R.sup.6a, R.sup.6b, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds of Formula (I) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.3, R.sup.4a, R.sup.4b, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds of Formula (II) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.5a, R.sup.5b, R.sup.5c, R.sup.5d, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the description of the compounds of Formula (III) provided herein. The following embodiments of moieties R.sup.1, R.sup.2, R.sup.6, R.sup.7, R.sup.8, R.sup.9, R.sup.10, R.sup.11, R.sup.12, m and n may be applied, alone or in combination, to the descriptions of the compounds of Formula (IV) provided herein.

[0164] R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0165] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein —O—(CH.sub.2).sub.m—C.sub.6-10aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O

—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and C.sub.6-10aryl is substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl

[0166] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkoxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl and thiophenyl; wherein alkyl, alkoxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0167] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; C.sub.2-4alkoxy; C.sub.4-6cycloalkyl; —O—C.sub.4-6cycloalkyl; phenyl; —O—(CH.sub.2).sub.m-phenyl; and thiophenyl; wherein phenyl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0168] In embodiments, R.sup.1 is selected from C.sub.2-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; C.sub.2-4alkoxy; C.sub.4-6cycloalkyl; —O—C.sub.4-6cycloalkyl; phenyl; —O—(CH.sub.2).sub.m-phenyl; and thiophenyl; wherein —O—(CH.sub.2).sub.m-phenyl is optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8 alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; and phenyl is substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3 alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3 and —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl.

[0169] In embodiments, R.sup.1 is selected from —CF.sub.2CF.sub.3; propyl; butyl; pentyl; propoxy; cyclobutyl; cyclohexyl; —O-cyclopentyl; thiophenyl; phenyl; —O-phenyl; —O—CH.sub.2-phenyl; wherein phenyl is optionally substituted with 1 or 2 substituents selected from —F, —Cl, —CH.sub.3, —O—(CH.sub.2).sub.5C≡CH, —O—(CH.sub.2).sub.7, —O—(CH.sub.2).sub.2C(N=N)(CH.sub.2).sub.2C≡CH, —O—(CH.sub.2).sub.2NHC(O)OC(CH.sub.3).sub.3, —O—CH.sub.2C≡CH, —O—(CH.sub.2).sub.5CF.sub.3 and —O—(CH.sub.2).sub.7.

[0170] In embodiments, R.sup.1 is selected from —CF.sub.2CF.sub.3; propyl; butyl; pentyl; propoxy; cyclobutyl; cyclohexyl; —O-cyclopentyl; thiophenyl; phenyl substituted with 1 or 2 substituents selected from —F, —Cl, —CH.sub.3, —O—(CH.sub.2).sub.5C≡CH, —O—(CH.sub.2).sub.7, —O—(CH.sub.2).sub.2C(N=N)(CH.sub.2).sub.2C≡CH, —O—(CH.sub.2).sub.2NHC(O)OC(CH.sub.3).sub.3, —O—CH.sub.2C≡CH, —O—(CH.sub.2).sub.5CF.sub.3 and —O—(CH.sub.2).sub.7.; —O-phenyl; —O—CH.sub.2-phenyl; wherein —O-phenyl and —O—CH.sub.2-phenyl is optionally substituted with 1 or 2 substituents selected from —F, —Cl, —CH.sub.3, —O—(CH.sub.2).sub.5C≡CH, —O—(CH.sub.2).sub.7, —O

—(CH.sub.2).sub.2C(N=N)(CH.sub.2).sub.2C≡CH, —O—(CH.sub.2).sub.2NHC(O)OC(CH.sub.3).sub.3, —O—CH.sub.2C≡CH, —O—(CH.sub.2).sub.5CF.sub.3 and —O—(CH.sub.2).sub.7.

[0171] In embodiments, R.sup.1 is cyclohexyl. In another embodiment, R.sup.1 is phenyl substituted with —F, —Cl, —CH.sub.3, —O—(CH.sub.2).sub.5C≡CH, —O—(CH.sub.2).sub.7, —O—(CH.sub.2).sub.2C(N=N)(CH.sub.2).sub.2C≡CH, —O—(CH.sub.2).sub.2NHC(O)OC(CH.sub.3).sub.3, —O—CH.sub.2C≡CH, —O—(CH.sub.2).sub.5CF.sub.3 and —O—(CH.sub.2).sub.7. In embodiments, R.sup.1 is phenyl.

[0172] R.sup.2 is selected from —H, -halo and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.2 is —H. In embodiments, R.sup.2 is -halo. In embodiments, R.sup.2 is —F. In embodiments, R.sup.2 is —C.sub.1-3alkyl. In embodiments, R.sup.2 is methyl.

[0173] R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; —C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups, for example 1 or 2 groups, selected from —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9.

[0174] In embodiments, R.sup.3 is selected from —H; —C.sub.24alkynyl; —C.sub.1-3alkyl optionally substituted with —C(O)NR.sup.8R.sup.9 or —NR.sup.8R.sup.9; and a 5 or 6 membered heterocycloalkyl optionally substituted with C.sub.1-3alkyl.

[0175] In embodiments, R.sup.3 is selected from —H; —C.sub.24alkynyl; —C.sub.1-3alkyl optionally substituted with —C(O)NR.sup.8R.sup.9 or —NR.sup.8R.sup.9; and a 5 or 6 membered nitrogen containing heterocycloalkyl optionally substituted with C.sub.1-3alkyl.

[0176] In embodiments, R.sup.3 is selected from —H; —C.sub.24alkynyl; —C.sub.1-3alkyl optionally substituted with —C(O)NR.sup.8R.sup.9 or —NR.sup.8R.sup.9; and piperidinyl optionally substituted with C.sub.1-3alkyl.

[0177] In embodiments, R.sup.3 is selected from methyl, ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2, —CH.sub.2C≡CH, —CH.sub.2C(O)N(CH.sub.3).sub.2 and N-methylpiperidine. In embodiments, R.sup.3 is selected from ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2, —CH.sub.2C≡CH, —CH.sub.2C(O)N(CH.sub.3).sub.2 and N-methylpiperidine.

[0178] In embodiments, R.sup.3 is selected from —C.sub.2-4alkynyl and —C.sub.1-3alkyl optionally substituted with —NR.sup.8R.sup.9.

[0179] In embodiments, R.sup.3 is selected from ethyl, i-propyl, —(CH.sub.2).sub.2N(CH.sub.3).sub.2, —(CH.sub.2).sub.3N(CH.sub.3).sub.2 and —CH.sub.2C≡CH.

[0180] In embodiments, R.sup.3 is i-propyl.

[0181] R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.4a is methyl and R.sup.4b is —H. In embodiments, R.sup.4a and R.sup.4b are both —H.

[0182] R.sup.4c and R.sup.4d are each independently selected from —H and C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d are each independently selected from —H and C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.4c and R.sup.4d are both —H or together with the carbon to which they are attached represent carbonyl. In embodiments, R.sup.4c and R.sup.4d are both —H.

[0183] R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and

—C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl. In embodiments, R.sup.5a is methyl and R.sup.5b, R.sup.5c and R.sup.5d are each —H. In embodiments, R.sup.5a, R.sup.5b and R.sup.5c are each —H and R.sup.5d is methyl. In embodiments, R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d each represent —H.

[0184] R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3alkyl. In embodiments, R.sup.6 is selected from —H; —Br; —NH.sub.2; —CN; methoxy; ethyl; —C(O)OCH.sub.3; —C(O)NH.sub.2; —C(O)OH; and —NHC(O)CH.sub.3.

[0185] R.sup.7 is selected from —NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-8cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0186] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocylic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; and C.sub.3-8cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0187] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocylic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN, methyl, ethyl, propyl, cyclopropyl, methoxy, —CH.sub.2CF.sub.3, —CH.sub.2OH, —CH.sub.2CH.sub.2OH, —C(O)OH, —(CH.sub.2).sub.2NHC(O)CH.sub.3 and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl.

[0188] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl and R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.pR.sup.12; a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl; a 5 or 6 membered monocyclic heteroaryl selected from pyridinyl, dihydropyranyl, oxazolyl, imidazolyl and thiazolyl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1 or 2 groups selected from —CN, methyl, ethyl, propyl, cyclopropyl, methoxy, —CH.sub.2CF.sub.3, —CH.sub.2OH, —CH.sub.2CH.sub.2OH, —C(O)OH, —(CH.sub.2).sub.2NHC(O)CH.sub.3 and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyranyl.

[0189] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl and R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12.

[0190] In embodiments, R.sup.7 is selected from NR.sup.10R.sup.11 wherein R.sup.10 is selected from methyl, ethyl or propyl; R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12; n is 1 or 2; and R.sup.12 is selected from isoxazolyl, oxadiazolyl, cyclopropyl, pyrazinyl, tetrahydrofuranyl and pyridinyl.

[0191] In embodiments, R.sup.7 is selected from a 5 to 7 membered monocyclic heterocycloalkyl optionally substituted with 1, 2 or 3 (for example, 1 or 2) groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl and C.sub.3-8-cycloalkyl; or the heterocycloakyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl.

[0192] In embodiments, R.sup.7 is a 5 to 7 membered monocyclic heterocycloalkyl selected from morpholinyl, thiazolidinyl, tetrahydropyranyl, pyrrolyl, thiomorpholinyl and 3,4-dihydro-2H-pyranyl wherein the heterocycloalkyl is optionally substituted with 1 or 2 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —Cl.sup.− 3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)OH; —CH.sub.2NHC(O)CH.sub.3; —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; and C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 6 membered monocyclic heterocycloalkyl.

[0193] In embodiments, R.sup.7 is a 5 to 7 membered monocyclic heterocyclalkyl optionally substituted with 1 or 2 substituents selected from methyl, ethyl, propyl, cyclopropyl, —CH.sub.2CH.sub.2OH, —CH.sub.2OH, —C(O)OH, —CH.sub.2CF.sub.3, and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyran.

[0194] In embodiments, R.sup.7 is morpholinyl optionally substituted with 1 or 2 substituents selected from methyl, ethyl, propyl, cyclopropyl, —CH.sub.2CH.sub.2OH, —CH.sub.2OH, —C(O)OH, —CH.sub.2CF.sub.3, and —CH.sub.2NHC(O)OC(CH.sub.3).sub.3; or optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form tetrahydropyran (i.e. R.sup.7 becomes a spirocyclic group).

[0195] In embodiments, R.sup.7 is 2-methylmorpholin-4-yl.

[0196] R.sup.8 is selected from —H and —C.sub.1-6alkyl. In embodiments, R.sup.8 is selected from —H and —C.sub.1-3alkyl. In embodiments, R.sup.8 is —H. In embodiments, R.sup.8 is —C.sub.1-3alkyl. In embodiments, R.sup.8 is methyl.

[0197] R.sup.9 is selected from —H and —C.sub.1-6alkyl. In embodiments, R.sup.9 is selected from —H and —C.sub.1-3alkyl. In embodiments, R.sup.9 is —H. In embodiments, R.sup.9 is —C.sub.1-3alkyl. In embodiments, R.sup.9 is methyl.

[0198] R.sup.10 is —C.sub.1-6alkyl. In embodiments, R.sup.10 is —C.sub.1-3alkyl. In embodiments, R.sup.10 is methyl. In another embodiment, R.sup.10 is ethyl. In embodiments, R.sup.10 is propyl.

[0199] R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3-alkoxy; or —(CH.sub.2).sub.nR.sup.12. In embodiments, R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and ethoxy. In embodiments, R.sup.11 is selected from ethyl, propyl, CH.sub.2CHF.sub.2, CH.sub.2CH.sub.2OCH.sub.2CH.sub.3 and —(CH.sub.2).sub.nR.sup.12. In embodiments, R.sup.1 is selected from —(CH.sub.2).sub.nR.sup.12.
[0200] R.sup.12 is selected from a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl. In embodiments, R.sup.12 is selected from isoxazolyl, oxadiazolyl, cyclopropyl, pyrazinyl, tetrahydrofuranyl and pyridinyl.
[0201] m is 0 or 1. In embodiments, m is 0. In embodiments, m is 1.
[0202] n is 1, 2 or 3. In embodiments, n is 1 or 2. In embodiments, n is 1. In embodiments, n is 2. In embodiments, n is 3.
[0203] In embodiments, the compound of Formula (I) is selected from: [0204] 2-(diethylamino)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0205] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0206] 6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-2-(1,3-thiazolidin-3-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0207] 2-[(2R,6S)-2,6-dimethylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0208] 6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-2-(thiomorpholin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0209] 2-[(2S)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0210] 2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0211] 2-[(2S,6S)-2,6-dimethylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0212] 2-(3-methylmorpholin-4-yl)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0213] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0214] 4-[(4-cyclohexylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0215] 4-[(4-cyclohexylphenyl)amino]-2-(2-methylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0216] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0217] 4-[(4-cyclohexylphenyl)amino]-2-((2R)-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0218] 4-[(4-cyclohexylphenyl)amino]-2-((2S)-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0219] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-[2-(2,2,2-trifluoroethyl)morpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0220] tert-butyl {[(2R)-4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholin-2-yl]methyl}carbamate; [0221] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-[2-(propan-2-yl)morpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0222] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(1,3-thiazolidin-3-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0223] 4-[(4-cyclohexylphenyl)amino]-2-[(2-ethoxyethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0224] 4-[(4-cyclohexylphenyl)amino]-2-(2-ethylmorpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0225] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(1,2-oxazol-3-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0226] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[2-(1,2,4-oxadiazol-3-yl)ethyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0227] 4-[(4-cyclohexylphenyl)amino]-2-(1,4-oxazepan-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0228] 4-[(4-cyclohexylphenyl)amino]-2-(1,9-dioxa-4-azaspiro[5.5]undecan-4-yl)-6-(propan-2-yl)-5,6-dihydro-

7H-pyrrolo[3,4-d]pyrimidin-7-one; [0229] 4-[(4-cyclohexylphenyl)amino]-2-(3-methoxypyrrolidin-1-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0230] 4-[(4-cyclohexylphenyl)amino]-2-[2-(2-hydroxyethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0231] 4-[(4-cyclohexylphenyl)amino]-2-(dipropylamino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0232] 4-[(4-cyclohexylphenyl)amino]-2-[(cyclopropylmethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0233] 4-[(4-cyclohexylphenyl)amino]-2-[2-(hydroxymethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0234] 4-[(4-cyclohexylphenyl)amino]-2-[3-(hydroxymethyl)morpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0235] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(pyrazin-2-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0236] 4-[(4-cyclohexylphenyl)amino]-2-(diethylamino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0237] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[(oxolan-2-yl)methyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0238] 4-[(4-cyclohexylphenyl)amino]-2-[(2,2-difluoroethyl)(methyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0239] 4-[(4-cyclohexylphenyl)amino]-2-{methyl[2-(pyridin-2-yl)ethyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0240] (3S)-4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholine-3-carboxylic acid; [0241] N-[2-(4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}morpholin-2-yl)ethyl]acetamide; [0242] 6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0243] 6-isopropyl-4-((4-isopropylphenyl)amino)-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one (Compound B); [0244] 4-{4-[(4-cyclohexylphenyl)amino]-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-2-yl}pyridine-2-carbonitrile; [0245] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0246] 4-[(4-cyclohexylphenyl)amino]-2-(2-methoxypyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0247] 4-[(4-cyclohexylphenyl)amino]-2-(2-methylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one (Compound C); [0248] 4-[(4-cyclohexylphenyl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0249] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0250] 4-[(4-cyclohexylphenyl)amino]-2-(1-methyl-1H-pyrazol-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0251] 4-[(4-cyclohexylphenyl)amino]-2-(1,3-oxazol-5-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0252] 4-[(4-cyclohexylphenyl)amino]-6-(propan-2-yl)-2-(1,3-thiazol-5-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0253] 2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0254] 4-{[4-(4-fluorophenoxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0255] 2-(2-cyclopropylmorpholin-4-yl)-4-({4′-[(hept-6-yn-1-yl)oxy][1,1′-biphenyl]-4-yl}amino)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0256] 2-(2-cyclopropylmorpholin-4-yl)-4-{[4′-(heptyloxy)[1,1′-biphenyl]-4-yl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0257] 4-[(4′-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1′-biphenyl]-4-yl)amino]-2-(2-cyclopropyl-morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0258] 2-[(2R)-2-methylmorpholin-4-yl]-4-[(4-pentylphenyl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0259] 4-{[4-(butan-2-yl)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0260] 4-{[4-(benzyloxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0261] 2-(2-cyclopropylmorpholin-4-yl)-4-{[4-(pentafluoroethyl)phenyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one;

[0262] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-[(4-propylphenyl)amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0263] 2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-[(4-propylphenyl)amino]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0264] 2-[(2R)-2-methylmorpholin-4-yl]-4-{[4-(pentafluoroethyl)phenyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0265] 2-(2-cyclopropylmorpholin-4-yl)-6-(propan-2-yl)-4-({4-[(propan-2-yl)oxy]phenyl}amino)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0266] 4-[(4-cyclobutylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0267] 4-{[4-(cyclopentyloxy)phenyl]amino}-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0268] 2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-4-{[4-(2,2,2-trifluoroethyl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0269] tert-butyl {2-[(4′-{[2-(2-cyclopropylmorpholin-4-yl)-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-yl]amino}[1,1′-biphenyl]-4-yl)oxy]ethyl}carbamate; [0270] 6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0271] 4-[(4-cyclohexylphenyl)amino]-6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0272] tert-butyl {2-[(4′-{[2-(morpholin-4-yl)-7-oxo-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-yl]amino}[1,1′-biphenyl]-4-yl)oxy]ethyl}carbamate; [0273] 4-[(4′-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1′-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0274] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-({4′-[(prop-2-yn-1-yl)oxy][1,1′-biphenyl]-4-yl}amino)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0275] 4-[(4-cyclohexylphenyl)amino]-6-[3-(dimethylamino)propyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0276] 4-[(4-cyclohexylphenyl)amino]-6-[2-(dimethylamino)ethyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0277] 4-[(4-cyclobutylphenyl)amino]-6-[3-(dimethylamino)propyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0278] 4-[(4-cyclobutylphenyl)amino]-6-[2-(dimethylamino)ethyl]-2-[(2R)-2-methylmorpholin-4-yl]-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0279] 2-(morpholin-4-yl)-4-{[4-(propan-2-yl)phenyl]amino}-6-(prop-2-yn-1-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0280] 4-[(4-cyclohexylphenyl)amino]-2-(oxan-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0281] 4-[(4-cyclohexylphenyl)amino]-2-(1H-imidazol-1-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0282] 2-(3,6-dihydro-2H-pyran-4-yl)-4-[(2′-methyl[1,1′-biphenyl]-4-yl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0283] 4-[(4′-{2-[3-(but-3-yn-1-yl)-3H-diaziren-3-yl]ethoxy}[1,1′-biphenyl]-4-yl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0284] 2-(3,6-dihydro-2H-pyran-4-yl)-4-[(2-fluoro[1,1′-biphenyl]-4-yl)amino]-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0285] 2-(morpholin-4-yl)-4-{[4-(pentafluoroethyl)phenyl]amino}-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0286] 4-[(2-fluoro[1,1′-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0287] 4-[(3′,4′-dichloro[1,1′-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0288] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-{[4-(propan-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0289] 4-[(4-tert-butylphenyl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0290] 4-[(2-methyl[1,1′-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0291] 4-[(4′-chloro[1,1′-biphenyl]-4-yl)amino]-2-(morpholin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0292] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0293] N-(4-cyclobutylphenyl)-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0294] N-(4-cyclohexylphenyl)-2-(2-cyclopropylmorpholin-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0295] 2-(2-cyclopropylmorpholin-4-yl)-N-[4′-(heptyloxy)[1,1′-biphenyl]-4-

yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0296] 2-[(2R)-2-methylmorpholin-4-yl]-N-{4′-[(6,6,6-trifluorohexyl)oxy][1,1′-biphenyl]-4-yl}-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0297] N-(4-cyclohexylphenyl)-2-(2-methylpyridin-4-yl)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine; [0298] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine (Compound D); [0299] 6-bromo-N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0300] N-(4-cyclohexylphenyl)-2-(3,6-dihydro-2H-pyran-4-yl)pyrido[2,3-d]pyrimidin-4-amine; [0301] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-8-oxo-8lambda-5--pyrido[2,3-d]pyrimidin-4-amine; [0302] N-(4-cyclohexylphenyl)-6-ethyl-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0303] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carbonitrile (Compound H); [0304] methyl 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxylate; [0305] methyl (R)-4-((4-cyclohexylphenyl)amino)-2-(2-methylmorpholino)pyrido[2,3-d]pyrimidine-6-carboxylate (Compound E); [0306] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxylic acid; [0307] 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carboxamide; [0308] 4-[(4-cyclohexylphenyl)amino]-2-(2-cyclopropylmorpholin-4-yl)pyrido[2,3-d]pyrimidine-6-carboxamide; [0309] N-(4-cyclohexylphenyl)-6-methoxy-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-4-amine; [0310] (R)—N-(4-cyclohexylphenyl)-6-methoxy-2-(2-methylmorpholino)pyrido[2,3-d]pyrimidin-4-amine (Compound F); [0311] N-{4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-6-yl}acetamide; [0312] N-4-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-4,6-diamine; [0313] N-(4-cyclohexylphenyl)-2-(morpholin-4-yl)-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0314] N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-6-(propan-2-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0315] 2-{4-[(4-cyclohexylphenyl)amino]-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydro-6H-pyrrolo[3,4-d]pyrimidin-6-yl}-N,N-dimethylacetamide (Compound G); [0316] N-(4-cyclohexylphenyl)-2-(2-cyclopropylmorpholin-4-yl)-6-(1-methylpiperidin-4-yl)-6,7-dihydro-5H-pyrrolo[3,4-d]pyrimidin-4-amine; [0317] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-{[4-(thiophen-2-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0318] 2-(morpholin-4-yl)-6-(propan-2-yl)-4-{[4-(thiophen-3-yl)phenyl]amino}-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one; [0319] and pharmaceutically acceptable salts thereof.

[0320] In embodiments, a compound of Formula (I) is:

##STR00023##

known as (R)—N-(4-cyclohexylphenyl)-2-(2-methylmorpholino)-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine (also referred herein to as N-(4-cyclohexylphenyl)-2-[(2R)-2-methylmorpholin-4-yl]-5,7-dihydrofuro[3,4-d]pyrimidin-4-amine, or also referred to herein as Compound A or CmpA), or a pharmaceutically acceptable salt thereof.

[0321] In embodiments, the compound of Formula (I) is Compound B:

##STR00024## [0322] 6-isopropyl-4-((4-isopropylphenyl)amino)-2-(pyridin-4-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one Compound B

[0323] In embodiments, the compound of Formula (I) is Compound C:

##STR00025## [0324] 4-[(4-cyclohexylphenyl)amino]-2-(2-methylpyridin-4-yl)-6-(propan-2-yl)-5,6-dihydro-7H-pyrrolo[3,4-d]pyrimidin-7-one Compound C

[0325] In embodiments, the compound of Formula (I) is Compound D: [0326] N-(4-cyclohexylphenyl

##STR00026## [2,3-d]pyrimidin-4-amine Compound D

[0327] In embodiments, the compound of Formula (I) is Compound E:

##STR00027## [0328] methyl (R)-4-((4-cyclohexylphenyl)amino)-2-(2-methylmorpholino) pyrido[2,3-d]pyrimidine-6-carboxylate Compound E

[0329] In embodiments, the compound of Formula (I) is Compound F:

##STR00028## [0330] (R)—N-(4-cyclohexylphenyl)-6-methoxy-2-(2-methylmorpholino) pyrido[2,3-d]pyrimidin-4-amine Compound F
[0331] In embodiments, the compound of Formula (I) is Compound G:
##STR00029## [0332] 2-(4-((4-cyclohexylphenyl)amino)-2-(3,6-dihydro-2H-pyran-4-yl)-5,7-dihydro-6H-pyrrolo[3,4-d]pyrimidin-6-yl)-N,N-dimethylacetamide Compound G
[0333] In embodiments, the compound according to Formula (I) is:
##STR00030##
known as (R)-4-((4-cyclohexylphenyl)amino)-2-(2-methylmorpholino)pyrido[2,3-d]pyrimidine-6-carbonitrile (also referred to as 4-[(4-cyclohexylphenyl)amino]-2-[(2R)-2-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidine-6-carbonitrile or Compound H), or a pharmaceutically acceptable salt thereof.
[0334] Atoms of the compounds and salts described herein may exist as their isotopes. All compounds of Formula (I) where an atom is replaced by one or more of its isotopes (for example a compound of Formula (I) where one or more carbon atom is an .sup.11C or .sup.13C carbon isotope, or where one or more hydrogen atoms is a .sup.2H or 3H isotope, or where one or more nitrogen atoms is a .sup.15N isotope or where one of more oxygen atoms is an 170 or 180 isotope) are encompassed herein.
[0335] Compounds herein may exist in one or more geometrical, optical, enantiomeric, and diastereomeric forms, including, but not limited to, cis- and trans-forms, E- and Z-forms, and R-, S- and meso-forms. Unless otherwise stated a reference to a particular compound includes all such isomeric forms, including racemic and other mixtures thereof. Where appropriate such isomers can be separated from their mixtures by the application or adaptation of known methods (e.g., chromatographic techniques and recrystallisation techniques). Where appropriate such isomers can be prepared by the application or adaptation of known methods. In embodiments, a single stereoisomer is obtained by isolating it from a mixture of isomers (e.g., a racemate) using, for example, chiral chromatographic separation. In embodiments, a single stereoisomer is obtained through direct synthesis from, for example, a chiral starting material.
[0336] In embodiments, there is provided a compound of Formula (I), or a pharmaceutically acceptable salt thereof, which is a single optical isomer being in an enantiomeric excess (% e.e.) of ≥95%, ≥98% or ≥99%. In embodiments, the single optical isomer is present in an enantiomeric excess (% e.e.) of ≥99%.
[0337] In embodiments, there is provided an N-oxide of a compound of Formula (I) as herein defined, or a pharmaceutically acceptable salt thereof.
[0338] Compounds of Formula (I), where R.sup.7 is —NR.sup.10R.sup.11 (i.e., R.sup.7 is linked by an aliphatic N atom), may for example be prepared as described in WO2021/180952 by the reaction of a compound of Formula (V):
##STR00031##
or a salt thereof, where R.sup.1, R.sup.2 and A are as defined herein, with an amine. The reaction is conveniently performed in a suitable solvent and at a suitable temperature, for example, di-isopropylethylamine in dimethylsulfoxide at a temperature of 20-100° C., or TsOH in butanol at 80° C.
[0339] When R.sup.7 is attached via a carbon atom, the compound of Formula (I) can be made by the reaction of a compound of Formula (V) with a boronic acid or ester of the Formula (VI), where R.sup.7 is as defined herein and each R is the same or different and represents —H, an aliphatic chain, or where together the two R groups form a ring with the boron and two oxygen atoms. The reaction is conveniently performed with a suitable base in the presence of a palladium catalyst and a solvent at a suitable temperature. For example, cesium carbonate or sodium carbonate and a palladium catalyst such as Pd(PPh.sub.3).sub.4, in aqueous dioxane at a temperature in the range of 80-100° C.
##STR00032##

[0340] When R.sup.7 is linked via an aromatic N atom, the compound of Formula (I) can be made by reaction of a compound of Formula (V) with the anion of R.sup.7. For example, by reaction of the anion of imidazole, generated by treatment with a suitable base (for example, sodium hydride), in a suitable solvent (for example dimethylformamide), with a compound of Formula (V).
[0341] A compound of the Formula (V) may be prepared from a compound of Formula (VII), or a salt thereof, where A is as defined herein, and a compound of Formula (VIII), or a salt thereof, where R.sup.1 and R.sup.2 are as defined herein, in the presence of a base in a suitable solvent (for example, di-isopropylethylamine in tert butanol or dimethylsulfoxide) and at a suitable temperature (for example 20-100° C.).
##STR00033##
[0342] A compound of Formula (I) may also be made in one pot from the reaction between a compound of Formula (VII) with the stepwise addition of a compound of Formula (VIII) and an amine R.sup.7. The reaction is conveniently performed in the presence of a base (for example, di-isopropylethylamine) in a suitable solvent (dimethylsulfoxide) at a suitable temperature (for example, a temperature of 20-100° C.).
[0343] A compound of Formula (VII) may be made, for example, from a compound of Formula (IX). Suitable conditions for this transformation are heating at a temperature of about 80° C. in POCl.sub.3 in the presence of an amine base such as diethylphenylamine.
##STR00034##
[0344] A compound of the Formula (IX) may, for example, be prepared from a compound of the Formula (X) by reaction with propan-2-amine and formaldehyde in a suitable solvent (for example, ethanol) at a suitable temperature (for example, a temperature of 0-80° C.).
##STR00035##
[0345] A compound of the Formula (I), when R.sup.5a and R.sup.5b are both H, may also be made from reaction of a compound of the Formula (XI), or a salt thereof, where R.sup.1, R.sup.2 and R.sup.7 are as defined herein, with a suitable amine, for example N, N-dimethylpropane-1,3-diamine. Suitable conditions for this reaction are HCl in ethanol at a temperature of about 190° C. in a sealed tube.
##STR00036##
[0346] It will be appreciated that certain of the various ring substituents in the compounds herein may be introduced by standard aromatic substitution reactions or generated by conventional functional group modifications either prior to or immediately following the processes mentioned above. For example, compounds of Formula (I) may be converted into further compounds of Formula (I) by conventional functional group modifications. Such reactions and modifications include, for example, introduction of a substituent by means of an aromatic substitution reaction, C—H activation reaction, reduction of substituents, alkylation of substituents and oxidation of substituents. The reagents and reaction conditions for such procedures are well known in the chemical art. Particular examples of aromatic substitution reactions include the introduction of a halogen group.
[0347] It will also be appreciated that in some of the reactions mentioned herein it may be necessary/desirable to protect any sensitive groups in the compounds. The instances where protection is necessary or desirable and suitable methods for protection are known to those skilled in the art. Conventional protecting groups may be used in accordance with standard practice (for illustration see T. W. Green, Protective Groups in Organic Synthesis, John Wiley and Sons, 1991). Thus, if reactants include groups such as amino, carboxy or hydroxy it may be desirable to protect the group in some of the reactions mentioned herein.
[0348] As mentioned previously, as a result of their KCC2 activation activity, the compounds of Formula (I), or pharmaceutically acceptable salts thereof, are useful in therapy, for example, in the treatment of neuropathic pain or neuro-inflammatory pain mediated at least in part by KCC2. Neuropathic pain caused by peripheral nerve damage, spinal cord injury, diabetic neuropathy, and

chemotherapy is associated with reduced KCC2 activity or increased NKCC1 activity in spinal dorsal horn neurons. Li et al., Cell Reports 15, 1376-1383, May 17, 2016. Without wishing to be bound by any particular theory, neuropathic pain or neuro-inflammatory pain can result from impaired Cl.sup.− transport. KCC2 is a K.sup.+—Cl.sup.− cotransporter and responsible for maintaining low Cl.sup.− concentration in neurons of the central nervous system (CNS), essential for postsynaptic inhibition through GABA.sub.A and glycine receptors. As a result of their KCC2 activation activity, the compounds of Formula (I), or pharmaceutically acceptable salts thereof, restore inhibition and normal function in patients experiencing neuropathic pain or neuro-inflammatory pain involving impaired Cl.sup.− transport, thereby relieving, alleviating or eliminating symptoms. In embodiments, the compounds of Formula (I), or pharmaceutically acceptable salts thereof, increase KCC2 activity, diminish neuronal hyperexcitabilty, and provide a GABAergic effect, resulting in beneficial therapeutic effects on neuropathic pain or neuro-inflammatory pain.

[0349] The term “therapy” is intended to have its normal meaning of dealing with a disease in order to entirely or partially relieve one, some or all of its symptoms, or to correct or compensate for the underlying pathology. In embodiments, the term “therapy” may also include “prophylaxis” if “prophylaxis” is specifically referred to. The term “prophylaxis” is intended to have its normal meaning and includes primary prophylaxis to prevent the development of the disease and secondary prophylaxis whereby the disease has already developed and the patient is temporarily or permanently protected against exacerbation or worsening of the disease or the development of new symptoms associated with the disease. Nonetheless, prophylactic (preventive) and therapeutic treatment are two separate embodiments of the disclosure herein. In embodiments, the term “treatment” can be used synonymously with “therapy”. Similarly, the term “treat” can be regarded as “applying therapy” where “therapy” is as defined herein. As used herein, the terms “treat”, “treatment” or “treating” as applied to neuropathic pain or neuro-inflammatory pain encompass any manner in which the symptoms or pathology of a condition, disorder or disease associated with neuropathic pain or neuro-inflammatory pain are ameliorated or otherwise beneficially altered. In embodiments, “treat”, “treatment” or “treating” can refer to inhibiting a disease or condition, e.g., arresting or reducing its development or at least one clinical or subclinical symptom thereof. In embodiments, “treat”, “treatment” or “treating” can refer to relieving the disease or condition, e.g., causing regression of the disease or condition or at least one of its clinical or subclinical symptoms. The terms “treat”, “treatment” or “treating” may be used interchangeably herein. The benefit to a subject being treated may be statistically significant, mathematically significant, or at least perceptible to the subject and/or the physician.

[0350] In embodiments, there is provided a compound of Formula (I), or a pharmaceutically acceptable salt thereof, for use in therapy for neuropathic pain or neuro-inflammatory pain. For example, in embodiments, Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt of any of the foregoing, is for use in therapy for neuropathic pain or neuro-inflammatory pain. In embodiments, there is provided the use of a compound of Formula (I), or a pharmaceutically acceptable salt thereof, e.g., Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt of any of the foregoing, for the manufacture of a medicament for neuropathic pain or neuro-inflammatory pain.

[0351] The term “therapeutically effective amount” refers to an amount of a compound of Formula (I) as described herein which is effective to provide “therapy” in a subject, or to “treat” a disease or disorder in a subject. In the case of neuropathic pain or neuro-inflammatory pain, the therapeutically effective amount may cause any of the changes observable or measurable in a subject as described in the definition of “therapy”, “treatment” and “prophylaxis” above. As recognized by those skilled in the art, effective amounts may vary depending on route of administration, excipient usage, and co-usage with other agents. For example, where a combination therapy is used, the amount of the compound of Formula (I) or pharmaceutically acceptable salt described in this specification and the amount of the other pharmaceutically active agent(s) are, when combined, jointly effective to treat a targeted disorder in the patient. In this context, the combined amounts are in a "therapeutically effective amount" if they are, when combined, sufficient to decrease the symptoms of neuropathic pain or neuro-inflammatory pain responsive to activation of KCC2 as described herein. Typically, such amounts may be determined by one skilled in the art by, for example, starting with the dosage range described in this specification for the compound of Formula (I) or pharmaceutically acceptable salt thereof and an approved or otherwise published dosage range(s) of the other pharmaceutically active compound(s). "Subjects" or "patients" include mammals, for example, humans. The terms "subjects" and "patients" may be used interchangeably herein.

[0352] The compounds of Formula (I), or pharmaceutically acceptable salts thereof, may be administered as pharmaceutical compositions, including one or more pharmaceutically acceptable excipients. Therefore, in embodiments, there is provided a pharmaceutical composition including a compound of Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.

[0353] The excipient(s) selected for inclusion in a particular composition will depend on factors such as the mode of administration and the form of the composition provided. Suitable pharmaceutically acceptable excipients are well known to persons skilled in the art and are described, for example, in the *Handbook of Pharmaceutical Excipients*, Sixth edition, Pharmaceutical Press, edited by Rowe, Ray C; Sheskey, Paul J; Quinn, Marian. Pharmaceutically acceptable excipients may function as, for example, adjuvants, diluents, carriers, stabilizers, flavorings, colorants, fillers, binders, disintegrants, lubricants, glidants, thickening agents and coating agents. As persons skilled in the art will appreciate, certain pharmaceutically acceptable excipients may serve more than one function and may serve alternative functions depending on how much of the excipient is present in the composition and what other excipients are present in the composition.

[0354] The pharmaceutical compositions may be in a form suitable for oral use (for example as tablets, lozenges, hard or soft capsules, films, dragees, aqueous or oily suspensions, emulsions, dispersible powders or granules, syrups or elixirs), for topical use (for example as creams, ointments, gels, lotions, transdermal patches, or aqueous or oily solutions or suspensions), for administration by inhalation (for example as a finely divided powder or a liquid aerosol), for administration by insufflation (for example as a finely divided powder) or for parenteral administration (for example as a sterile aqueous or oily solution or suspension for intravenous, subcutaneous or intramuscular dosing), or as a suppository for rectal dosing. The compositions may be obtained by conventional procedures using conventional pharmaceutical excipients well known in the art. Thus, compositions intended for oral use may contain, for example, one or more coloring, sweetening, flavoring and/or preservative agents.

[0355] In embodiments, a pharmaceutical composition herein may contain a compound according to Formula 1 in any of the amounts set forth herein and a diluent in an amount from 30% to 90%. Examples of diluents include lactose, microcrystalline cellulose, starch, calcium phosphate, calcium carbonate, sucrose, mannitol, maltodextrin and sorbitol. In embodiments, a pharmaceutical composition herein may contain a compound according to Formula 1 and a lubricant in an amount, e.g., from 0.25% to 5.0%. Examples of lubricants include magnesium stearate, stearic acid, sodium stearyl fumarate, talc, polyethylene glycols and silicon dioxide. Examples of water-soluble lubricants include sodium benzoate, polyethylene glycol, and adipic acid. In embodiments, a pharmaceutical composition herein may contain a compound according to Formula 1 and a disintegrant in an amount, e.g., from 1.0% to 10.0%. There are two classes of disintegrants: traditional disintegrants, such as starch, and super disintegrants, which include croscarmellose sodium, crospovidone, and sodium starch glycolate.

[0356] The compound of Formula (I) will normally be administered to a subject, e.g., a warm-blooded animal at a unit dose within the range 2.5-5000 mg/m.sup.2 body area of the animal, or approximately 0.05-100 mg/kg, and this normally provides a therapeutically-effective dose. A unit dose form such as a tablet, capsule, film, patch, vial will can contain, for example 0.1-500 mg of active ingredient. The daily dose will necessarily be varied depending upon the host treated, the particular route of administration, any therapies being co-administered, and the severity of the illness being treated.
[0357] The pharmaceutical compositions described herein include compounds of Formula (I), or a pharmaceutically acceptable salt thereof, for use in therapy for neuropathic pain or neuro-inflammatory pain.
[0358] As such, in embodiments, there is provided a pharmaceutical composition for use in therapy for neuropathic pain or neuro-inflammatory pain, including a compound of Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.
[0359] In embodiments, there is provided a pharmaceutical composition for use in the treatment of neuropathic pain or neuro-inflammatory pain in which activation of KCC2 is beneficial, including a compound of Formula (I), or a pharmaceutically acceptable salt thereof, and at least one pharmaceutically acceptable excipient.
[0360] In embodiments, treatment of neuropathic pain or neuro-inflammatory pain is implemented by administering to a subject, e.g., a human, in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain, about 0.01 mg to about 1500 mg of Formula (I), or a pharmaceutically acceptable salt thereof, such as Compound A, Compound B, Compound C, Compound D, Compound E, Compound F, Compound G, or Compound H, or a pharmaceutically acceptable salt of any of the foregoing. In embodiments, methods include treating neuropathic pain or neuro-inflammatory pain by administering to a subject, e.g., a human, in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain about 0.01 mg to about 1500 mg of Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, the amount Formula (I), or a pharmaceutically acceptable salt thereof, can be, e.g., between 0.1 and 1500 mg/day, or 0.01 mg/kg/day to 15 mg/kg/day, for treatment of neuropathic pain or neuro-inflammatory pain. For example, the daily dosage can be, e.g., in the range of about 0.01 to 1500 mg, 0.1 to 1250 mg, 0.1 to 1000 mg, 0.1 to 750 mg, 0.1 to 500 mg, 0.1 to 450 mg, 0.1 to 300 mg, 0.1 to 250 mg, 0.1 to 200 mg, 0.1 to 175 mg, 0.1 to 150 mg, 0.1 to 125 mg, 0.1 to 100 mg, 0.1 to 75 mg, 0.1 to 50 mg, 0.1 to 30 mg, 0.1 to 25 mg, 0.1 to 20 mg, 0.1 to 15 mg, 0.1 to 10 mg, 0.1 to 5 mg, 0.1 to 1 mg, 1 to 1500 mg, 1 to 1000 mg, 1 to 500 mg, 1 to 300 mg, 1 to 250 mg, 1 to 200 mg, 1 to 175 mg, 1 to 150 mg, 1 to 125 mg, 1 to 100 mg, 1 to 75 mg, 1 to 50 mg, 1 to 30 mg, 1 to 25 mg, 1 to 20 mg, 1 to 15 mg, 1 to 10 mg, 1 to 5 mg, 5 to 1500 mg, 5 to 1000 mg, 5 to 500 mg, 5 to 300 mg, 5 to 250 mg, 5 to 200 mg, 5 to 175 mg, 5 to 150 mg, 5 to 125 mg, 5 to 100 mg, 5 to 75 mg, 5 to 50 mg, 5 to 30 mg, 5 to 25 mg, 5 to 20 mg, 5 to 15 mg, 5 to 10 mg, 10 to 1500 mg, 10 to 1000 mg, 10 to 500 mg, 10 to 300 mg, 10 to 250 mg, 10 to 200 mg, 10 to 175 mg, 10 to 150 mg, 10 to 125 mg, 10 to 100 mg, 10 to 75 mg, 10 to 50 mg, 10 to 30 mg, 10 to 25 mg, 10 to 20 mg, 10 to 15 mg, 15 to 1500 mg, 15 to 1000 mg, 15 to 500 mg, 15 to 300 mg, 15 to 250 mg, 15 to 200 mg, 15 to 175 mg, 15 to 150 mg, 15 to 125 mg, 15 to 100 mg, 15 to 75 mg, 15 to 50 mg, 15 to 30 mg, 15 to 25 mg, 15 to 20 mg, 20 to 1500 mg, 20 to 1000 mg, 20 to 500 mg, 20 to 300 mg, 20 to 250 mg, 20 to 200 mg, 20 to 175 mg, 20 to 150 mg, 20 to 125 mg, 20 to 100 mg, 20 to 75 mg, 20 to 50 mg, 20 to 30 mg, 20 to 25 mg, 25 to 1500 mg, 25 to 1000 mg, 25 to 500 mg, 25 to 300 mg, 25 to 250 mg, 25 to 200 mg, 25 to 175 mg, 25 to 150 mg, 25 to 125 mg, 25 to 100 mg, 25 to 75 mg, 25 to 50 mg, 25 to 30 mg, 30 to 1500 mg, 30 to 1000 mg, 30 to 500 mg, 30 to 300 mg, 30 to 250 mg, 30 to 200 mg, 30 to 175 mg, 30 to 150 mg, 30 to 125 mg, 30 to 100 mg, 30 to 75 mg, 30 to 50 mg, 35 to 1500 mg, 35 to 1000 mg, 35 to 500 mg, 35 to 300 mg, 35 to 250 mg, 35 to 200 mg, 35 to 175 mg, 35 to 150 mg, 35 to 125 mg, 35 to 100 mg, 35 to 75 mg, 35 to 50 mg, 40 to 1500 mg, 40 to 1000 mg, 40 to 500 mg, 40 to 300 mg, 40 to 250 mg, 40 to 200 mg, 40 to 175 mg, 40 to 150 mg, 40 to 125 mg, 40 to 100 mg, 40 to 75

mg, 40 to 50 mg, 50 to 1500 mg, 50 to 1000 mg, 50 to 500 mg, 50 to 300 mg, 50 to 250 mg, 50 to 200 mg, 50 to 175 mg, 50 to 150 mg, 50 to 125 mg, 50 to 100 mg, 50 to 75 mg, 75 to 1500 mg, 75 to 1000 mg, 75 to 500 mg, 75 to 300 mg, 75 to 250 mg, 75 to 200 mg, 75 to 175 mg, 75 to 150 mg, 75 to 125 mg, 75 to 100 mg, 100 to 1500 mg, 100 to 1000 mg, 100 to 500 mg, 100 to 300 mg, 100 to 250 mg, 100 to 200 mg, 100 to 175 mg, 100 to 150 mg, 100 to 125 mg, 125 to 1500 mg, 125 to 1000 mg, 125 to 500 mg, 125 to 300 mg, 125 to 250 mg, 125 to 200 mg, 125 to 175 mg, 125 to 150 mg, 150 to 1500 mg, 150 to 1000 mg, 150 to 500 mg, 150 to 300 mg, 150 to 250 mg, 150 to 200 mg, 150 to 175 mg, 175 to 1500 mg, 175 to 1000 mg, 175 to 500 mg, 175 to 300 mg, 175 to 250 mg, 175 to 200 mg, 200 to 1500 mg, 200 to 1000 mg, 200 to 500 mg, 200 to 300 mg, 200 to 250 mg, 250 to 1500 mg, 250 to 1000 mg, 250 to 500 mg, 250 to 300 mg, 7.5 to 15 mg, 2.5 to 5 mg, 1 to 5 mg, with doses of, e.g., about 0.25 mg, 0.5 mg, 0.75 mg, 1 mg, 1.25 mg, 1.5 mg, 1.75 mg, 2.0 mg, 2.5 mg, 3.0 mg, 3.5 mg, 4.0 mg, 4.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg, 15 mg, 17.5 mg, 20 mg, 22.5 mg, 25 mg, 27.5 mg, 30 mg, 35 mg, 40 mg, 45 mg, 50 mg, 75 mg, 100 mg, 125 mg, 150 mg, 175 mg, 200 mg, 225 mg, 250 mg, 275 mg, 300 mg, 400 mg and 500 mg being examples.

[0361] In embodiments, pharmaceutical compositions for treating neuropathic pain or neuro-inflammatory pain may include Formula (I), or a pharmaceutically acceptable salt thereof, in an amount of, e.g., about 0.01 to 1500 mg, 0.01 to 1250 mg, 0.01 to 1000 mg, 0.01 to 750 mg, 0.01 to 500 mg, 0.01 to 250 mg, 0.01 to 100 mg, 0.01 to 50 mg, 0.01 to 25 mg, 0.01 to 10 mg, 0.01 to 5 mg, 0.01 to 1 mg, 0.1 to 500 mg, 0.1 to 450 mg, 0.1 to 300 mg, 0.1 to 250 mg, 0.1 to 200 mg, 0.1 to 175 mg, 0.1 to 150 mg, 0.1 to 125 mg, 0.1 to 100 mg, 0.1 to 75 mg, 0.1 to 50 mg, 0.1 to 30 mg, 0.1 to 25 mg, 0.1 to 20 mg, 0.1 to 15 mg, 0.1 to 10 mg, 0.1 to 5 mg, 0.1 to 1 mg, 0.5 to 500 mg, 0.5 to 450 mg, 0.5 to 300 mg, 0.5 to 250 mg, 0.5 to 200 mg, 0.5 to 175 mg, 0.5 to 150 mg, 0.5 to 125 mg, 0.5 to 100 mg, 0.5 to 75 mg, 0.5 to 50 mg, 0.5 to 30 mg, 0.5 to 25 mg, 0.5 to 20 mg, 0.5 to 15 mg, 0.5 to 10 mg, 0.5 to 5 mg, 0.5 to 1 mg, 1 to 500 mg, 1 to 450 mg, 1 to 300 mg, 1 to 250 mg, 1 to 200 mg, 1 to 175 mg, 1 to 150 mg, 1 to 125 mg, 1 to 100 mg, 1 to 75 mg, 1 to 50 mg, 1 to 30 mg, 1 to 25 mg, 1 to 20 mg, 1 to 15 mg, 1 to 10 mg, 1 to 5 mg, 5 to 500 mg, 5 to 450 mg, 5 to 300 mg, 5 to 250 mg, 5 to 200 mg, 5 to 175 mg, 5 to 150 mg, 5 to 125 mg, 5 to 100 mg, 5 to 75 mg, 5 to 50 mg, 5 to 30 mg, 5 to 25 mg, 5 to 20 mg, 5 to 15 mg, 5 to 10 mg, 10 to 500 mg, 10 to 450 mg, 10 to 300 mg, 10 to 250 mg, 10 to 200 mg, 10 to 175 mg, 10 to 150 mg, 10 to 125 mg, 10 to 100 mg, 10 to 75 mg, 10 to 50 mg, 10 to 30 mg, 10 to 25 mg, 10 to 20 mg, 10 to 15 mg, 15 to 500 mg, 15 to 450 mg, 15 to 300 mg, 15 to 250 mg, 15 to 200 mg, 15 to 175 mg, 15 to 150 mg, 15 to 125 mg, 15 to 100 mg, 15 to 75 mg, 15 to 50 mg, 15 to 30 mg, 15 to 25 mg, 15 to 20 mg, 20 to 500 mg, 20 to 450 mg, 20 to 300 mg, 20 to 250 mg, 20 to 200 mg, 20 to 175 mg, 20 to 150 mg, 20 to 125 mg, 20 to 100 mg, 20 to 75 mg, 20 to 50 mg, 20 to 30 mg, 20 to 25 mg, 25 to 500 mg, 25 to 450 mg, 25 to 300 mg, 25 to 250 mg, 25 to 200 mg, 25 to 175 mg, 25 to 150 mg, 25 to 125 mg, 25 to 100 mg, 25 to 80 mg, 25 to 75 mg, 25 to 50 mg, 25 to 30 mg, 30 to 500 mg, 30 to 450 mg, 30 to 300 mg, 30 to 250 mg, 30 to 200 mg, 30 to 175 mg, 30 to 150 mg, 30 to 125 mg, 30 to 100 mg, 30 to 75 mg, 30 to 50 mg, 40 to 500 mg, 40 to 450 mg, 40 to 400 mg, 40 to 250 mg, 40 to 200 mg, 40 to 175 mg, 40 to 150 mg, 40 to 125 mg, 40 to 100 mg, 40 to 75 mg, 40 to 50 mg, 50 to 500 mg, 50 to 450 mg, 50 to 300 mg, 50 to 250 mg, 50 to 200 mg, 50 to 175 mg, 50 to 150 mg, 50 to 125 mg, 50 to 100 mg, 50 to 75 mg, 75 to 500 mg, 75 to 450 mg, 75 to 300 mg, 75 to 250 mg, 75 to 200 mg, 75 to 175 mg, 75 to 150 mg, 75 to 125 mg, 75 to 100 mg, 100 to 500 mg, 100 to 450 mg, 100 to 300 mg, 100 to 250 mg, 100 to 200 mg, 100 to 175 mg, 100 to 150 mg, 100 to 125 mg, 125 to 500 mg, 125 to 450 mg, 125 to 300 mg, 125 to 250 mg, 125 to 200 mg, 125 to 175 mg, 125 to 150 mg, 150 to 500 mg, 150 to 450 mg, 150 to 300 mg, 150 to 250 mg, 150 to 200 mg, 200 to 500 mg, 200 to 450 mg, 200 to 300 mg, 200 to 250 mg, 250 to 500 mg, 250 to 450 mg, 250 to 300 mg, 300 to 500 mg, 300 to 450 mg, 300 to 400 mg, 300 to 350 mg, 350 to 500 mg, 350 to 450 mg, 350 to 400 mg, 400 to 500 mg, 400 to 450 mg, with 0.1 mg, 0.25 mg, 0.5 mg, 0.75 mg, 1 mg, 2.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg, 15 mg, 17.5 mg, 20 mg, 22.5 mg, 25 mg, 30 mg, 35 mg, 40 mg, 45 mg, 50 mg, 55 mg, 60 mg, 65 mg, 70 mg, 75 mg, 80 mg, 85 mg, 90 mg, 95 mg, 100 mg, 125 mg, 150 mg 175 mg, 200 mg, 225 mg, 250 mg, 275 mg, 300 mg, 325 mg, 350 mg, 375 mg, 400 mg, 425 mg, 450 mg, 475 mg, and 500 mg being examples.

[0362] Typically, dosages may be administered to a subject experiencing neuropathic pain or neuro-inflammatory pain once, twice, three or four times daily, every other day, once weekly, or once a month. In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered to a subject experiencing neuropathic pain or neuro-inflammatory pain once a day (e.g., morning or night), or twice a day, (e.g., morning and evening), three times a day (e.g., at breakfast, lunch, and dinner, or every 8 hours), or four times a day (e.g., breakfast, lunch, dinner and at bedtime) at a dose of 0.01-1000 mg/administration. In embodiments, the pharmaceutical compositions described herein may be administered by continuous infusion.

[0363] In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered to a subject experiencing neuropathic pain or neuro-inflammatory pain 1500 mg/per day, 1400 mg/per day, 1300 mg/per day, 1200 mg/per day, 1000 mg/per day, 900 mg/per day, 800 mg/per day, 700 mg/per day, 600 mg/per day, 500 mg/per day, 400 mg/per day, 300 mg/per day, 200 mg/per day, 100 mg/per day, 95 mg/per day, 90 mg/per day, 85 mg/per day, 80 mg/per day, 75 mg/per day, 70 mg/per day, 65 mg/per day, 60 mg/per day, 55 mg/per day, 50 mg/per day, 45 mg/per day, 40 mg/per day, 35 mg/per day, 30 mg/per day, 25 mg/per day, 20 mg/per day, 15 mg/per day, 10 mg/per day, 5 mg/per day, 4 mg/per day, 3 mg/per day, 3 mg/per day, 2 mg/per day, 1 mg/per day, in one or more doses. Dosages can be lower for infants and children than for adults. In embodiments, an infant or pediatric dose can be about 0.1 to 1500 mg per day once or in 2, 3 or 4 divided doses. In embodiments, a pediatric dose can be 0.05 mg/kg/day to 1500 mg/kg/day. In embodiments, the subject may be started at a low dose and the dosage is escalated over time. It should be understood that the above amounts are exemplary and doses of Formula (I), or a pharmaceutically acceptable salt thereof, can include different amounts and varying ranges within a continuum between the minimum or maximum amounts described above in connection with Formula (I), or a pharmaceutically acceptable salt thereof.

[0364] In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 1 hour after administration to the subject. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including (Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 2 hours after administration to the subject. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 3 hours after administration to the subject. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 4 hours after administration to the subject. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 6 hours after administration to the subject. In embodiments, methods of treating neuropathic pain or neuro-inflammatory pain are provided which include administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 8, 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration to the subject. In embodiments, improvement in at least one symptom for 12 hours after administration of the pharmaceutical composition to the subject is provided in accordance with the present disclosure. In embodiments, the pharmaceutical compositions provide improvement in next day functioning of the subject. For example, the pharmaceutical compositions may provide improvement in one or more symptoms of neuropathic pain or neuro-inflammatory pain for more than about, e.g., 2 hours, 4 hours, 6 hours, 8 hours, 10 hours, 12 hours, 14 hours, 16 hours, 18 hours, 20 hours, 22 hours or 24 hours after administration and waking from a night of sleep.

[0365] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, in an effective amount, after an early sign of one or more symptoms of neuropathic pain or neuro-inflammatory pain is detected to reduce or prevent further symptoms. In embodiments, a continuing regimen of administration of active agents herein according to Formula (I), or a pharmaceutically acceptable salt thereof, is effective to reduce or prevent occurrence of symptoms associated with neuropathic pain or neuro-inflammatory pain.

[0366] In embodiments, the methods described herein are effective to reduce, delay, or prevent one or more other clinical symptoms of neuropathic pain or neuro-inflammatory pain. For example, the effect of a composition including Formula (I), or a pharmaceutically acceptable salt thereof, on a particular symptom, pharmacologic, or physiologic indicator can be compared to an untreated subject, or the condition of the subject prior to treatment. In embodiments, the symptom, pharmacologic, and/or physiologic indicator is measured in a subject prior to treatment, and again one or more times after treatment is initiated. In embodiments, the control is a reference level, or average determined based on measuring the symptom, pharmacologic, or physiologic indicator in one or more subjects that do not have the disease or condition to be treated (e.g., healthy subjects). In embodiments, the effect of the treatment is compared to a conventional treatment that is known the art.

[0367] Indeed, clinical efficacy of treatment can be monitored using any method known in the art. Measurable parameters to monitor efficacy will depend on the condition being treated. For monitoring the status or improvement of pain, various pain assessment scales and tools are known. For example, the numeric pain rating scale (NRS), typically ranging from one to ten with one being the least severe and 10 being the most severe, can be used. Another example is the visual analogue scale (VAS), which is a measurement instrument that tries to measure a characteristic or attitude that is believed to range across a continuum of values and cannot easily be directly measured. For example, the amount of pain that a patient feels ranges across a continuum from none to an extreme amount of pain.

[0368] Effective treatment of neuropathic pain or neuro-inflammatory pain herein may be established by showing reduction in the frequency or severity of symptoms (e.g., more than, e.g., 10%, 20%, 30% 40%, 50% or more) after a period of time compared with baseline. For example, after a baseline period of 1 month, the subjects may be randomly allocated Formula (I), or a pharmaceutically acceptable salt thereof, or placebo as add-on therapy to standard therapies, during a double-blind period of, e.g., 2 months.

[0369] In embodiments, primary outcome measurements may include the percentage of responders on Formula (I), or a pharmaceutically acceptable salt thereof, and on placebo, defined as having experienced at least a 10% to 50% or more reduction of symptoms during the second month of the double-blind period compared with baseline.

[0370] Cognitive impairment (aka impairment of cognition) may be associated with subjects experiencing neuropathic pain or neuro-inflammatory pain. Cognitive impairment may be measured against normal cognitive function, which refers to the normal physiologic activity of the brain, including, but not limited to, one or more of the following: mental stability, memory/recall abilities, problem solving abilities, reasoning abilities, thinking abilities, judging abilities, ability to discriminate or make choices, capacity for learning, ease of learning, perception, intuition, attention, and awareness, as measured by any criteria suitable in the art.

[0371] Cognitive impairment also includes deficits in mental activities that are mild or that otherwise do not significantly interfere with daily life. Mild cognitive impairment (MCI) is an example of such a condition. A subject with mild cognitive impairment may display symptoms of dementia as described above (e.g., difficulties with language or memory) but the severity of these symptoms is such that a diagnosis of dementia may not be appropriate.

[0372] One skilled in the art will appreciate that there are numerous human and animal models that may be used to evaluate and compare the relative safety and efficacy of compounds described herein for the treatment of cognitive impairment associated with neuropathic pain or neuro-inflammatory pain. In humans, cognitive function may be measured, for example and without limitation, by the clinical global impression of change scale (CGI); the Mini Mental State Exam (MMSE) (aka the Folstein Test); the Neuropsychiatric Inventory (NPI); the Clinical Dementia Rating Scale (CDR); the Cambridge Neuropsychological Test Automated Battery (CANTAB), the Sandoz Clinical Assessment-Geriatric (SCAG) scale, the Benton Visual Retention Test (BVRT), Montreal Cognitive Assessment (MoCA) or Digit Symbol Substitution Test (DSST).

[0373] In animal model systems, cognitive function may be measured in various conventional ways known in the art, including using a Morris Water Navigation Task, Barnes maze, radial arm maze task, T maze and the like. Other tests known in the art may also be used to assess cognitive function, such as novel object recognition and odor recognition tasks.

[0374] Cognitive function may also be measured using imaging techniques such as Positron Emission Tomography (PET), functional magnetic resonance imaging (fMRI), Single Photon Emission Computed Tomography (SPECT), or any other imaging technique that allows one to measure brain function. In animals, cognitive function may also be measured with electrophysiological techniques.

[0375] In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is administered via a pharmaceutical composition for treatment of neuropathic pain or neuro-inflammatory pain. Pharmaceutical compositions herein encompass dosage forms. Dosage forms herein encompass unit doses. In embodiments, as discussed below, various dosage forms including conventional formulations and modified release formulations can be administered one or more times daily. Any suitable route of administration may be utilized, e.g., enteral or parenteral including oral, rectal, nasal, pulmonary, vaginal, sublingual, transdermal, intravenous, intraarterial, intramuscular, intraperitoneal and subcutaneous routes. As mentioned previously, suitable dosage forms include tablets, capsules, oral liquids, powders, aerosols, transdermal modalities such as topical liquids, patches, creams and ointments, parenteral formulations and suppositories.

[0376] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 50% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0377] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 55% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 55% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0378] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 60% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 60% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration, in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0379] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 65% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration. In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain Formula (I), or a pharmaceutically acceptable salt thereof, which provides an in vivo plasma profile, wherein the in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, in the subject 10 hours after administration of (Formula (I), or a pharmaceutically acceptable salt thereof, is reduced by more than 65% and the method provides improvement in the subject for more than 10, 12, 14, 16, 18, 20, 22 or 24 hours after administration.

[0380] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain wherein the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is less than about 75% of the administered dose. In embodiments, provided herein are methods wherein the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is less than about 75%.

[0381] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain wherein the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is less than about 80% of the administered dose. In embodiments, provided herein are methods wherein the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is less than about 80% of the administered dose.

[0382] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain wherein the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject about 4 hours after administration of the pharmaceutical composition is between about 65% to about 85% of the administered dose. In embodiments, the amount of Formula (I), or a pharmaceutically acceptable salt thereof, within the subject after about, e.g., 6 hours, 8 hours, 10 hours, 12 hours, 15 hours, or 20 hours after administration of the pharmaceutical composition is between about 65% to about 85% of the administered dose.

[0383] In embodiments, as mentioned previously, pharmaceutical compositions herein may be provided with conventional release or modified release profiles. Pharmaceutical compositions may be prepared using a pharmaceutically acceptable “carrier” or “excipient” composed of materials that are considered safe and effective. The “carrier” includes all components present in the pharmaceutical formulation other than the active ingredient or ingredients. The term “carrier” includes, but is not limited to, diluents, binders, lubricants, disintegrants, fillers, and coating compositions. Those with skill in the art are familiar with such pharmaceutical carriers and methods of compounding pharmaceutical compositions using such carriers.

[0384] In embodiments, pharmaceutical compositions herein are modified release dosage forms which provide modified release profiles. Modified release profiles may exhibit immediate release, delayed release, or extended release profiles. Conventional (or unmodified) release oral dosage forms such as tablets, capsules, suppositories, syrups, solutions and suspensions typically release medications into the mouth, stomach or intestines as the tablet, capsule shell or suppository dissolves, or, in the case of syrups, solutions and suspensions, when they are swallowed. The pattern of drug release from modified release (MR) dosage forms is deliberately changed from that of a conventional dosage form to achieve a desired therapeutic objective and/or better patient compliance. Types of MR drug products include orally disintegrating dosage forms (ODDFs) which provide immediate release, extended release dosage forms, delayed release dosage forms (e.g., enteric coated), and pulsatile release dosage forms.

[0385] An ODDF is a solid dosage form containing a medicinal substance or active ingredient which disintegrates rapidly, usually within a matter of seconds when placed upon the tongue. The disintegration time for ODDFs generally range from one or two seconds to about a minute. ODDFs are designed to disintegrate or dissolve rapidly on contact with saliva. This mode of administration can be beneficial to people who may have problems swallowing tablets whether it be from physical infirmity or psychiatric in nature. Subjects with neuropathic pain or neuro-inflammatory pain may exhibit such behavior. ODDF's can provide rapid delivery of medication to the blood stream through mucosa resulting in a rapid onset of action. Examples of ODDFs include orally disintegrating tablets, capsules and rapidly dissolving films and wafers.

[0386] Extended release dosage forms (ERDFs) have extended release profiles and are those that allow a reduction in dosing frequency as compared to that presented by a conventional dosage form, e.g., a solution or unmodified release dosage form. ERDFs provide a sustained duration of action of a drug. Suitable formulations which provide extended release profiles are well-known in the art. For example, coated slow release beads or granules ("beads" and "granules" are used interchangeably herein) in which Formula (I), or a pharmaceutically acceptable salt thereof, is applied to beads, e.g., confectioners nonpareil beads, and then coated with conventional release retarding materials such as waxes, enteric coatings and the like. In embodiments, beads can be formed in which Formula (I), or a pharmaceutically acceptable salt thereof, is mixed with a material to provide a mass from which the drug leaches out. In embodiments, the beads may be engineered to provide different rates of release by varying characteristics of the coating or mass, e.g., thickness, porosity, using different materials, etc. Beads having different rates of release may be combined into a single dosage form to provide variable or continuous release. The beads can be contained in capsules or compressed into tablets.

[0387] In embodiments, modified dosage forms herein incorporate delayed release dosage forms having delayed release profiles. Delayed release dosage forms can include delayed release tablets or delayed release capsules. A delayed release tablet is a solid dosage form which releases a drug (or drugs) such as Formula (I), or a pharmaceutically acceptable salt thereof, at a time other than promptly after administration. A delayed release capsule is a solid dosage form in which the drug is enclosed within either a hard or soft soluble container made from a suitable form of gelatin, and which releases a drug (or drugs) at a time other than promptly after administration. For example, enteric-coated tablets, capsules, particles and beads are well-known examples of delayed release dosage forms. Enteric coated tablets, capsules and particles and beads pass through the stomach and release the drug in the intestine. In embodiments, a delayed release tablet is a solid dosage form containing a conglomerate of medicinal particles that releases a drug (or drugs) at a time other than promptly after administration. In embodiments, the conglomerate of medicinal particles are covered with a coating which delays release of the drug. In embodiments, a delayed release capsule is a solid dosage form containing a conglomerate of medicinal particles that releases a drug (or drugs) at a time other than promptly after administration. In embodiments, the conglomerate of medicinal particles is covered with a coating which delays release of the drug.

[0388] Delayed release dosage forms are known to those skilled in the art. For example, coated delayed release beads or granules in which one Formula (I), or a pharmaceutically acceptable salt thereof, is applied to beads, e.g., confectioners nonpareil beads, and then coated with conventional release delaying materials such as waxes, enteric coatings and the like. In embodiments, beads can be formed in which Formula (I), or a pharmaceutically acceptable salt thereof, is mixed with a material to provide a mass from which the drug leaches out. In embodiments, the beads may be engineered to provide different rates of release by varying characteristics of the coating or mass, e.g., thickness, porosity, using different materials, etc. In embodiments, enteric coated granules of Formula (I), or a pharmaceutically acceptable salt thereof, can be contained in an enterically coated capsule or tablet which releases the granules in the small intestine. In embodiments, the granules have a coating which remains intact until the coated granules reach at least the ileum and thereafter provide a delayed release of the drug in the colon. Suitable enteric coating materials are well known in the art, e.g., Eudragit® coatings such methacrylic acid and methyl methacrylate polymers and others. The granules can be contained in capsules or compressed into tablets.

[0389] In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is incorporated into porous inert carriers that provide delayed release profiles. In embodiments, the porous inert carriers incorporate channels or passages from which the drug diffuses into surrounding fluids. In embodiments, Formula (I), or a pharmaceutically acceptable salt thereof, is incorporated into an ion-exchange resin to provide a delayed release profile. Delayed action may result from a predetermined rate of release of the drug from the resin when the drug-resin complex contacts gastrointestinal fluids and the ionic constituents dissolved therein. In embodiments, membranes are utilized to control rate of release from drug containing reservoirs. In embodiments, liquid preparations may also be utilized to provide a delayed release profile. For example, a liquid preparation consisting of solid particles dispersed throughout a liquid phase in which the particles are not soluble. The suspension is formulated to allow at least a reduction in dosing frequency as compared to that drug presented as a conventional dosage form (e.g., as a solution or a prompt drug-releasing, conventional solid dosage form). For example, a suspension of ion-exchange resin constituents or microbeads.

[0390] In embodiments, pharmaceutical compositions described herein are suitable for parenteral administration, including, e.g., intramuscular (i.m.), intravenous (i.v.), subcutaneous (s.c.), intraperitoneal (i.p.), or intrathecal (i.t.). Parenteral compositions must be sterile for administration by injection, infusion or implantation into the body and may be packaged in either single-dose or multi-dose containers. In embodiments, liquid pharmaceutical compositions for parenteral administration to a subject include Formula (I), or a pharmaceutically acceptable salt thereof, in any of the respective amounts described above. In embodiments, the pharmaceutical compositions for parenteral administration are formulated as a total volume of about, e.g., 10 ml, 20 ml, 25 ml, 50 ml, 100 ml, 200 ml, 250 ml, or 500 ml. In embodiments, the compositions are contained in a bag, a glass vial, a plastic vial, or a bottle.

[0391] In embodiments, pharmaceutical compositions for parenteral administration include respective amounts described above for Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, pharmaceutical compositions for parenteral administration include about 0.05 mg to about 500 mg Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, pharmaceutical compositions for parenteral administration to a subject include Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of about 0.005 mg/ml to about 500 mg/ml. In embodiments, the pharmaceutical composition for parenteral administration includes Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of, e.g., about 0.05 mg/ml to about 50 mg/ml, about 0.1 mg/ml to about 50 mg/ml, about 0.1 mg/ml to about 10 mg/ml, about 0.05 mg/ml to about 25 mg/ml, about 0.05 mg/ml to about 10 mg/ml, about 0.05 mg/ml to about 5 mg/ml, or about 0.05 mg/ml to about 1 mg/ml. In embodiments, the pharmaceutical composition for parenteral administration includes Formula (I), or a pharmaceutically acceptable salt thereof, at a respective concentration of, e.g., about 0.05 mg/ml to about 15 mg/ml, about 0.5 mg/ml to about 10 mg/ml, about 0.25 mg/ml to about 5 mg/ml, about 0.5 mg/ml to about 7 mg/ml, about 1 mg/ml to about 10 mg/ml, about 5 mg/ml to about 10 mg/ml, or about 5 mg/ml to about 15 mg/ml.

[0392] In embodiments, a pharmaceutical composition for parenteral administration is provided wherein the pharmaceutical composition is stable for at least six months. In embodiments, the pharmaceutical compositions for parenteral administration exhibit no more than about 5% decrease in Formula (I), or a pharmaceutically acceptable salt thereof, e.g., 3 months or 6 months. In embodiments, the amount of Formula (I), or a pharmaceutically acceptable salt thereof, degrades at no more than about, e.g., 2.5%, 1%, 0.5% or 0.1%. In embodiments, the degradation is less than about, e.g., 5%, 2.5%, 1%, 0.5%, 0.25%, 0.1%, for at least six months.

[0393] In embodiments, pharmaceutical compositions for parenteral administration are provided wherein the pharmaceutical composition remains soluble. In embodiments, pharmaceutical compositions for parenteral administration are provided that are stable, soluble, local site compatible and/or ready-to-use. In embodiments, the pharmaceutical compositions herein are ready-to-use for direct administration to a subject in need thereof.

[0394] The pharmaceutical compositions for parenteral administration provided herein may include one or more excipients, e.g., solvents, solubilty enhancers, suspending agents, buffering agents, isotonicity agents, stabilizers or antimicrobial preservatives. When used, the excipients of the parenteral compositions will not adversely affect the stability, bioavailability, safety, and/or efficacy of Formula (I), or a pharmaceutically acceptable salt thereof, used in the composition. Thus, parenteral compositions are provided wherein there is no incompatibility between any of the components of the dosage form.

[0395] In embodiments, parenteral compositions including (Formula (I), or a pharmaceutically acceptable salt thereof, include a stabilizing amount of at least one excipient. For example, excipients may be selected from the group consisting of buffering agents, solubilizing agents, tonicity agents, antioxidants, chelating agents, antimicrobial agents, and preservative. One skilled in the art will appreciate that an excipient may have more than one function and be classified in one or more defined group.

[0396] In embodiments, parenteral compositions including Formula (I), or a pharmaceutically acceptable salt thereof, and an excipient wherein the excipient is present at a weight percent (w/v) of less than about, e.g., 10%, 5%, 2.5%, 1%, or 0.5%. In embodiments, the excipient is present at a weight percent between about, e.g., 1.0% to 10%, 10% to 25%, 15% to 35%, 0.5% to 5%, 0.001% to 1%, 0.01% to 1%, 0.1% to 1%, or 0.5% to 1%. In embodiments, the excipient is present at a weight percent between about, e.g., 0.001% to 1%, 0.01% to 1%, 1.0% to 5%, 10% to 15%, or 1% to 15%.

[0397] In embodiments, parenteral compositions may be administered as needed, e.g., once, twice, thrice or four or more times daily, or continuously depending on the subject's needs.

[0398] In embodiments, parenteral compositions of Formula (I), or a pharmaceutically acceptable salt thereof, are provided, wherein the pH of the composition is between about 4.0 to about 8.0. In embodiments, the pH of the compositions is between, e.g., about 5.0 to about 8.0, about 6.0 to about 8.0, about 6.5 to about 8.0. In embodiments, the pH of the compositions is between, e.g., about 6.5 to about 7.5, about 7.0 to about 7.8, about 7.2 to about 7.8, or about 7.3 to about 7.6. In embodiments, the pH of the aqueous solution is, e.g., about 6.8, about 7.0, about 7.2, about 7.4, about 7.6, about 7.7, about 7.8, about 8.0, about 8.2, about 8.4, or about 8.6.

[0399] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition including Formula (I), or a pharmaceutically acceptable salt thereof, in a respective amount described herein, wherein the composition provides an in vivo plasma profile having a C.sub.max less than about 800 ng/ml. In embodiments, the composition provides improvement for more than 6 hours after administration to the subject.

[0400] In embodiments, pharmaceutical compositions including Formula (I), or a pharmaceutically acceptable salt thereof, provide an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a C.sub.max less than about, e.g., 2000 ng/ml, 1000 ng/ml, 850 ng/ml, 800 ng/ml, 750 ng/ml, 700 ng/ml, 650 ng/ml, 600 ng/ml, 550 ng/ml, 450 ng/ml, 400 ng/ml 350 ng/ml, or 300 ng/ml and wherein the composition provides improvement of next day functioning of the subject. In embodiments, the pharmaceutical composition provides an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a C.sub.max less than about, e.g., 250 ng/ml, 200 ng/ml 150 ng/ml, or 100 ng/ml and wherein the composition provides improvement of next day functioning of the subject. In embodiments, the pharmaceutical composition provides improvement in one or more symptoms of neuropathic pain or neuro-inflammatory pain for more than 6 hours after administration.

[0401] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, wherein the composition provides a consistent in vivo plasma profile having a AUC.sub.0-∞ of less than about 900 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides improvement in next day functioning of the subject. In embodiments, the compositions provide an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 850 ng.Math.hr/ml, 800 ng.Math.hr/ml, 750 ng.Math.hr/ml, or 700 ng.Math.hr/ml and wherein the pharmaceutical composition provides improvement of next day functioning of the subject. In embodiments, the composition provides improvement in neuropathic pain or neuro-inflammatory pain for more than 6 hours after administration.

[0402] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a pharmaceutical composition comprising Formula (I), or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical composition provides an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 650 ng.Math.hr/ml, 600 ng.Math.hr/ml, 550 ng.Math.hr/ml, 500 ng.Math.hr/ml, or 450 ng.Math.hr/ml. In embodiments, the composition provides an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 400 ng.Math.hr/ml, 350 ng.Math.hr/ml, 300 ng.Math.hr/ml, 250 ng.Math.hr/ml, or 200 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides an in vivo plasma profile of Formula (I), or a pharmaceutically acceptable salt thereof, having a AUC.sub.0-∞ of less than about, e.g., 150 ng.Math.hr/ml, 100 ng.Math.hr/ml, 75 ng.Math.hr/ml, or 50 ng.Math.hr/ml. In embodiments, the pharmaceutical composition provides improvement of next day functioning of the subject after administration for more than, e.g., 4 hours, 6 hours, 8 hours, 10 hours, or 12 hours, after administration of the composition to the subject.

[0403] In embodiments, the T.sub.max provided by a pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 3 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 2.5 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 2 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 1.5 hours. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 1 hour. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 0.5 hour. In embodiments, the T.sub.max provided by the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, is less than 0.25 hour.

[0404] In embodiments, the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least about 80% within the first 20 minutes of administration to a subject in need thereof. In embodiments, the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least about, e.g., 85%, 90% or 95% within the first 20 minutes of administration to a subject in need thereof. In embodiments, the pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, provides a dissolution of at least 80% within the first 10 minutes of administration to a subject in need thereof.

[0405] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a first pharmaceutical dosage including a sub-therapeutic amount of Formula (I), or a pharmaceutically acceptable salt thereof. In embodiments, treating neuropathic pain or neuro-inflammatory pain includes administering to a subject in need thereof a pharmaceutical composition containing Formula (I), or a pharmaceutically acceptable salt thereof, in a sub-therapeutic amount, wherein the composition provides improvement in one or more symptoms of the neuropathic pain or neuro-inflammatory pain for more than 6 hours after administration.

[0406] A sub-therapeutic dosage is an amount of Formula (I), or a pharmaceutically acceptable salt thereof, that is less than the amount typically required for a therapeutic effect. In embodiments, a sub-therapeutic dosage is an amount of Formula (I), or a pharmaceutically acceptable salt thereof, that alone may not provide improvement in at least one symptom of neuropathic pain or neuro-inflammatory pain but is sufficient to maintain such improvement. In embodiments, the methods provide administering a first pharmaceutical composition containing an effective amount of Formula (I), or a pharmaceutically acceptable salt thereof, that provides improvement in at least one symptom of neuropathic pain or neuro-inflammatory pain and a second composition containing a subtherapeutic amount of Formula (I), or a pharmaceutically acceptable salt thereof, that maintains the improvement. In embodiments, after administration of the first pharmaceutical composition, the second pharmaceutical composition may provide a synergistic effect to improve at least one symptom of neuropathic pain or neuro-inflammatory pain.

[0407] In embodiments, provided herein are methods of treating neuropathic pain or neuro-inflammatory pain including administering to a subject in need thereof or diagnosed with neuropathic pain or neuro-inflammatory pain a first pharmaceutical composition including a first pharmaceutical dosage of Formula (I), or a pharmaceutically acceptable salt thereof, wherein the first pharmaceutical dosage provides improvement for more than 6 hours after administration, and a second pharmaceutical composition including a sub-therapeutic dosage of Formula (I), or a pharmaceutically acceptable salt thereof.

[0408] In embodiments, the first or the second pharmaceutical composition are provided to the subject once in the evening and once in the morning. In embodiments, the total amount of Formula (I), or a pharmaceutically acceptable salt thereof, administered to a subject in a 24-hour period is any of the respective amounts described herein.

[0409] In embodiments, the first and/or the second pharmaceutical compositions may be provided with conventional release or modified release profiles. The first and second pharmaceutical compositions may be provided at the same time or separated by an interval of time, e.g., 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 12 hours, etc. In embodiments, the first and the second pharmaceutical compositions may be provided with different drug release profiles to create a two-phase release profile. For example, the first pharmaceutical composition may be provided with an immediate release profile, e.g., ODDF, parenteral, etc., and the second pharmaceutical composition may provide an extended release profile. In embodiments, one or both of the first and second pharmaceutical compositions may be provided with an extended release or delayed release profile. Such compositions may be provided as pulsatile formulations, multilayer tablets or capsules containing tablets, beads, granules, etc. In embodiments, the first pharmaceutical composition is an immediate release composition. In embodiments, the second pharmaceutical composition is an immediate release composition. In embodiments, the first and second pharmaceutical compositions are provided as separate immediate release compositions, e.g., film, tablets or capsules. In embodiments the first and second pharmaceutical compositions are provided 12 hours apart.

[0410] It should be understood that respective dosage amounts of (Formula (I), or a pharmaceutically acceptable salt thereof, that are provided herein are applicable to all the dosage forms described herein including conventional dosage forms, modified dosage forms, the first and second pharmaceutical compositions, as well as the parenteral formulations described herein. Those skilled in the art will determine appropriate amounts depending on criteria such as dosage form, route of administration, subject tolerance, efficacy, therapeutic goal and therapeutic benefit, among other pharmaceutically acceptable criteria.

[0411] Combination therapies utilizing Formula (I), or a pharmaceutically acceptable salt thereof, can include administration of the active agents together in the same admixture, or in separate admixtures. In embodiments, the pharmaceutical composition can include two, three, or more active agents. In embodiments, the combinations result in a more than additive effect on the treatment of the disease or disorder. Thus, treatment is provided for neuropathic pain or neuro-inflammatory pain with a combination of agents that combined, may provide a synergistic effect that enhances efficacy.

[0412] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosure herein belongs.

[0413] The term "about" or "approximately" as used herein means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, "about" can mean within 3 or more than 3 standard deviations, per the practice in the art. Alternatively, "about" can mean a range of up to 20%, up to 10%, up to 5%, and/or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value.
[0414] "Improvement" refers to the treatment of neuropathic pain or neuro-inflammatory pain from any cause, measured relative to at least one symptom of neuropathic pain or neuro-inflammatory pai.
[0415] "Improvement in next day functioning" or "wherein there is improvement in next day functioning" refers to improvement after waking from an overnight sleep period wherein the beneficial effect of administration of (Formula (I), or a pharmaceutically acceptable salt thereof, applies to at least one symptom of a syndrome or disorder herein and is discernable, either subjectively by a subject or objectively by an observer, for a period of time, e.g., 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 12 hours, 24 hours, etc. after waking.
[0416] "PK" refers to the pharmacokinetic profile. C.sub.max is defined as the highest plasma drug concentration estimated during an experiment (ng/ml). T.sub.max is defined as the time when C.sub.max is estimated (min). AUC.sub.0-∞ is the total area under the plasma drug concentration-time curve, from drug administration until the drug is eliminated (ng.Math.hr/ml or μg.Math.hr/ml). The area under the curve is governed by clearance. Clearance is defined as the volume of blood or plasma that is totally cleared of its content of drug per unit time (ml/min).
[0417] "Pharmaceutically acceptable" refers to molecular entities and compositions that are "generally regarded as safe", e.g., that are physiologically tolerable and do not typically produce an allergic or similar untoward reaction, such as gastric upset and the like, when administered to a human. In embodiments, this term refers to molecular entities and compositions approved by a regulatory agency of the federal or a state government, as the GRAS list under section 204(s) and 409 of the Federal Food, Drug and Cosmetic Act, that is subject to premarket review and approval by the FDA or similar lists, the U.S. Pharmacopeia or another generally recognized pharmacopeia for use in animals, and more particularly in humans.
[0418] "Effective amount" or "therapeutically effective amount", previously referred to, can also mean a dosage sufficient to alleviate one or more symptoms of a syndrome, disorder, disease, or condition being treated, or to otherwise provide a desired pharmacological and/or physiologic effect. "Effective amount" or "therapeutically effective amount" may be used interchangeably herein.
[0419] "Co-administered with", "administered in combination with", "a combination of", "in combination with" or "administered along with" may be used interchangeably and mean that two or more agents are administered in the course of therapy. The agents may be administered together at the same time or separately in spaced apart intervals. The agents may be administered in a single dosage form or in separate dosage forms.
[0420] "Subject in need thereof" may include individuals that have been diagnosed with neuropathic pain or neuro-inflammatory pain from any cause. The methods may be provided to any individual including, e.g., wherein the subject is a neonate, infant, a pediatric subject (6 months to 12 years), an adolescent subject (age 12-18 years) or an adult (over 18 years). Subjects include mammals. "Subject" and "patient" are used interchangeably herein.
[0421] "Prodrug" refers to a pharmacological substance (drug) that is administered to a subject in an inactive (or significantly less active) form. Once administered, the prodrug is metabolized in the body (in vivo) into a compound having the desired pharmacological activity.
[0422] "Analog" and "Derivative" may be used interchangeably and refer to a compound that

possesses the same core as the parent compound, but may differ from the parent compound in bond order, the absence or presence of one or more atoms and/or groups of atoms, and combinations thereof. The derivative can differ from the parent compound, for example, in one or more substituents present on the core, which may include one or more atoms, functional groups, or substructures. In general, a derivative can be imagined to be formed, at least theoretically, from the parent compound via chemical and/or physical processes.

[0423] The term “pharmaceutically acceptable salt”, as used herein, refers to derivatives of the compounds defined herein, wherein the parent compound is modified by making acid or base salts thereof. Example of pharmaceutically acceptable salts include but are not limited to mineral or organic acid salts of basic residues such as amines; and alkali or organic salts of acidic residues such as carboxylic acids. The pharmaceutically acceptable salts include the conventional non-toxic salts or the quaternary ammonium salts of the parent compound formed, for example, from non-toxic inorganic or organic acids. Such conventional non-toxic salts include but are not limited to those derived from inorganic acids such as hydrochloric, hydrobromic, sulfuric, sulfamic, phosphoric, and nitric acids; and the salts prepared from organic acids such as acetic, propionic, succinic, glycolic, stearic, lactic, malic, tartaric, citric, ascorbic, pamoic, maleic, hydroxymaleic, phenylacetic, glutamic, benzoic, salicylic, sulfanilic, 2-acetoxybenzoic, fumaric, tolunesulfonic, naphthalenesulfonic, methanesulfonic, ethane disulfonic, oxalic, and isethionic salts. The pharmaceutically acceptable salts can be synthesized from the parent compound, which contains a basic or acidic moiety, by conventional chemical methods.

EXAMPLES

[0424] The examples provided herein are included solely for augmenting the disclosure herein and should not be considered to be limiting in any respect.

[0425] The following Examples establish: Compound A is capable of directly binding to KCC2, as measured using cellular thermal shift assay (CETSA). Compound A potently increases KCC2 activity without modifying its plasma membrane stability or the phosphorylation of key regulatory sites, as measured by patch-clamp recording and TI flux. This contrasts to other indirect potentiators of KCC2, which have been shown to increase expression levels on the plasma membrane, enhance its expression, increase transcription of the SLC12A5 gene, or modify its phosphorylation (Gagnon et al., Nat. Med. 19, 1524-1528 (2013); Lee et al., Brain: a journal of neurology 145, 950-963 (2022); Prael Iii et al.,. Front Cell Dev Biol 10, 912812 (2022); Tang et al., Sci. Transl. Med. 11. 10.1126) (2019)). Consistent with its role in limiting neuronal excitability, Compound A reduced neuronal Cl.sup.− accumulation and the development of late recurrent discharges (LRDs) in acute brain slices exposed to 0-Mg. Compound A rapidly accumulated in the brain when dosed IV, IP, or SC to concentrations ranging from 40-0.6 μM, dependent upon the route of administration. Importantly, as measured in the open field, Compound A appeared to have no gross effects on mouse behavior.

[0426] The Examples establish that in mice treated with Compound A and diazepam, neuronal cell death in the hippocampus was significantly reduced compared to controls treated with diazepam alone. This effect may be due to the ability of Compound A to limit hyperexcitability, but previous in vitro and in vivo studies have shown that reducing KCC2 expression levels, or transiently inhibiting it, are sufficient to induce neuronal apoptosis. Likewise, modifying KCC2 expression levels during development also impacts neuronal viability.

Example 1

Identification of Small Molecules that Activate and Directly Bind to KCC2

[0427] A multi-tiered high-throughput screening assay was used, leveraging a proprietary library of 1.3 million small molecule compounds from AstraZeneca (Cambridge, UK) to identify chemical entities capable of activating KCC2. A single point concentration of 30 μM was used to identify potassium flux using an established thallium (TI+) influx FLIPR assay as the primary read out in HEK-293 cells overexpressing KCC2 (Cardarelli et al., Nat. Med. 23, 1394-1396 (2017); Conway et al., J Biol Chem 292, 21253-21263 (2017); Lee et al., 2022, supra). Compounds that exhibited >20% potentiation of KCC2 activity with EC_{50} of <30 μM were selected for further optimization. This approach resulted in the identified subseries of fused pyrimidine compounds as KCC2 activators, which are denoted as subseries A (SSA; FIG. **1**A).

[0428] Neuronal Cultures. Hippocampal neurons were prepared from Sprague Dawley E18 embryos and plated at 450,000 per dish and maintained at 37° C. in a humidified 5% CO_2 incubator for 10-25 days before experimentation (Lee et al., Nat. Neurosci. 14, 736-743 (2011)).

[0429] Cell Culture and biotinylation. HEK-293 cells were purchased from ATCC (CRL-1573) and transfected with human KCC2 and GFP expression plasmids using Lipofectamine 2000. Cells were utilized for experimentation 48-72 h following transfection. Cells were biotinylated at 4° C. using NHS-Biotin. Following lysis and purification on immobilized avidin beads, cell surface and total fractions were subject to immunoblotting as outlined previously (Conway et al., 2017, supra).

[0430] FLIPR TI Flux assay. HEK-293 cells were washed with HBSS, and then incubated with loading buffer containing TI sensitive fluorescent dye (https://www.moleculardevices.com), 20 μM bumetanide 20 μM ouabain and 2.5 mM probenecid for 1 h at room temperature (RT) under control conditions or with drugs in the dark using a FlexStation 3 microplate reader (Molecular Devices) and imaged 525 nM for 30 min. Stimulus buffer was added containing 5 mM K^+/0.5 mM Tl^+ and cells were imaged every 1.5 over a 190 second time course The initial rate Tl^+ uptake (Relative Fluorescence Units (RFU) per second) was determined following addition of the stimulus buffer as outlined previously (Conway et al., 2017, supra) (Lee et al., 2022, supra).

Example 2

Identification of Compound A

[0431] Determining if compounds in SSA directly interact with KCC2. This was tested using a cellular thermal shift assay (CETSA), which measures the ability of a drug to modify the thermal stability of its target in whole cells (Kawatkar et al., ACS Chem. Biol. 14, 1913-1920 (2019); Martinez et al., Sci Rep 8, 9472 (2018)).

[0432] 2×10^6 transfected HEK-293 cells were incubated with 30 μM drug or vehicle for 90 min at 37° C. and then heated from 37-58° C. for 5 min. Cells were rapidly chilled to 25° C. and NP40 was added to a final concentration of prior to snap freezing in liquid nitrogen. Following 5 freeze/thaw cycles lysates were centrifuged at 12,000×g to remove denatured proteins and associated cell debris. The supernatant cell and tissue lysate samples, or immunoprecipitated KCC2 samples were separated by SDS-PAGE and subsequently immunoblotted with KCC2 antibodies (1:2000, Millipore #07-432), KCC2-S940, KCC2-pT1007(1:2000, PhosphosSolutions, #1568 actin and GAPDH (1:3000, SantaCruz #sc-32233) as detailed in previous studies (Lee et al., 2011, supra) (Conway et al., 2017, supra), and the level of soluble KCC2 at each temperature was normalized to that seen at 37° C. This data was then used to determine the temperature at which 50% of KCC2 was denatured.

[0433] Aliquots of HEK-293 cells expressing, KCC2 were exposed to 30 μM SSA, heated from 37-55° C. The remaining level of soluble native KCC2 at each temperature was measured by immunoblotting and normalized to the level seen at 37° C. which was given a value of 1. SSA-1 reduced the thermal stability of KCC2 compared to vehicle, which demonstrated target engagement. To gain insights into affinity, isothermal dose response analysis was performed at 50° C. revealing a K_D of 395.6±23.5 nM of SSA for KCC2 (FIG. **1**B-D). SSA derivatives were further optimized for affinity, optimal drug like properties and predicted brain penetration by in vitro cellular uptake assays, which resulted in the identification of Compound A (FIG. **1** E). Dose response measurements with Compound A were then made at 50° C. and this data was used to estimate its relative affinity for KCC2 (Kawatkar er al., 2019, supra) (Martinez et al., 2018, supra). To control for compounds that may indirectly modulate KCC2 activity, via modifying its phosphorylation (Kahle et al., Trends Neurosci. 36, 726-737 (2013); Moore et al., Proc. Natl. Acad. Sci. U.S.A. 115, 10166-10171 (2018); Silayeva et al., Proc. Natl. Acad. Sci. U.S.A 112, 3523-3528

(2015); Smalley et al., Front. Mol. Neurosci. 13, 563091 (2020)), the effect of Compound A to modify the activity of a panel of 144 protein kinases (SelectScreen; Thermofisher) was studied. This panel included protein kinase C isoforms, OSR1, SPAK (STK39) and with-no-lysine kinases (WNK), all of which phosphorylate KCC2 on multiple residues to indirectly modify its activity (Kahle et al., 2013, supra; Smalley et al., 2020, supra). Compound A did not modify the activity of this kinase panel.

[0434] As measured by TI+ flux, Compound A exhibited an EC.sub.50 for KCC2 of 261.4±22.2 nM and a maximum potentiation of 90% at a saturating concentration of 3 μM using a 1% β-cyclodextrin (BCD) vehicle (FIG. **1**F). Several studies have shown that the activity and plasma membrane accumulation of KCC2 is potentiated via phosphorylation of S940, a process mediated by the activity of protein kinase C and protein phosphatase 1 (Lee et al, 2011, supra; Lee et al., J Biol Chem 282, 29777-29784 (2007); Moore et al., Proc. Natl. Acad. Sci. U.S.A 115, 10166-10171 (2018); Moore et al., Trends Neurosci. 40, 555-571 (2017); Silayeva et al., Proc. Natl. Acad. Sci. U.S.A. 112, 3523-3528 (2015)). The ability of Compound A to modulate the activity of a KCC2 construct was tested in which this key residue for phospho-dependent modulation of KCC2 was mutated to an alanine (S940A). Significantly, Compound A increased the activity of KCC2-S940A construct (FIG. **1**G). Finally, the specificity of Compound A for other solute transporters was examined using the TI+ assay using HEK-293 cells expressing KCC3, KCC4, or NKCC1. Compared to KCC2, effects on KCC3, KCC4, and NKCC1 were only seen at concentrations more than 35-fold that of KCC2 (>100 μM) (FIG. **1**G).

[0435] These results suggest that Compound A binds directly to KCC2 and acts to increase its activity with mnthimal effects on other solute transporters that regulate neuronal Cl.sup.− accumulation.

Example 3

Compound a Potentiates KCC2 Activity without Modifying its Plasma Membrane Accumulation or Phosphorylation

[0436] To confirm the validity of our measurements using TI+flux, KCC2 was expressed in HEK-293 cells with the al subunit of the glycine receptor (GlyRα1) which forms homomeric glycine-activated ion channels when expressed in this system. The gramicidin perforated patch clamp technique was used to measure the reversal potential for GlyRα1-mediated currents (E.sub.GLY) in cells exposed to Compound A. The use of this technique, while limited in throughput, facilitates high resolution single-cell recordings without disturbing endogenous intracellular Cl.sup.− levels (Cardarelli et al., 2017, supra; Conway et al., 2017, supra). To calculate E.sub.GLY, cells were exposed to 50 μM Gly and the polarity of Gly-induced currents was determined at different holding potentials. Brief (15 min) incubation with either 0.3 μM (approx. EC.sub.50) or 3 μM Compound A induced shifts of −15.6±4.5 (p=0.022), and −32.2±9.5 (p=0.001) mV in E.sub.GLY respectively, while vehicle was without effect (−0.8±0.5 mv; p=0.512) (FIG. **2**A-B). As determined using the Nernst equation, these negative shifts in E.sub.GLY by Compound A corresponded to reduced intracellular Cl.sup.− levels of FIG. **2**B; Compound A; 5.2±1.2; vehicle=7.9±1.2 mM; p=0.0104).

[0437] The effects of Compound A on the plasma membrane stability of KCC2 expression was assessed using biotinylation, as described previously (Cardarelli et al., 2017, supra; Conway et al., 2017, supra). Compared to vehicle, 0.3 μM or 3 μM Compound A did not modify the level of KCC2 on the plasma membrane (FIG. **2**C; 94.5 ±8.4%, p=0.745 and 104.7±8.4% of control; p=0.736 respectively). The reliability of the biotinylation procedure was assessed via immunoblotting with antibodies against the cytosolic protein actin. Actin was present in the total lysates but was absent from the surface fractions (FIG. **2**C).

[0438] The activity of KCC2 is subject to dynamic modulation via phosphorylation of critical intracellular residues within its C-terminal cytoplasmic domain. Central to this process are serine 940 (S940) and threonine 1007 (T1007), which activate and inhibit KCC2 activity respectively (Moore et al., 2017, supra). Therefore, the effects of Compound A on the phosphorylation of these residues was examined using immunoblotting with characterized phospho-specific antibodies pS940 and pT1007, respectively. Compound A (3 μM) did not significantly modify phosphorylation of S940 (95.3±10.2% of control; p=0.764), or T1007 (108±10.2% of control; p=0.649) (FIG. **2**D).

[0439] The foregoing suggests that Compound A potentiates KCC2 activity without modifying either its plasma membrane accumulation or the phosphorylation of key regulatory sites.

Example 4

[0440] Compound A reduces neuronal Cl.sup.− accumulation in vitro

[0441] To test if Compound A modifies neuronal Cl.sup.− levels gramicidin-perforated patch-clamp recording was used to measure the reversal potentials of GABA.sub.A receptor mediated currents (E.sub.GABA) in 18-20 Div hippocampal neurons (Lee et al., 2011, supra; Moore et al., 2018, supra). At this developmental stage, GABA.sub.AR receptor activation leads to hyperpolarization, a phenomenon critically dependent upon KCC2 (Kontou et al., J Biol Chem, 100364 (2021)).

[0442] Neuronal Patch clamp assays. Neuronal cell culture recordings were performed in bath saline at 34° C. For gramicidin perforated-patch experiments, pipettes contained (in mM): 140 KCl and 10 HEPES, pH 7.4 KOH. For whole-cell experiments, pipettes contained (in mM): 115 K-meth-SO.sub.4, 30 KCl, 2 Mg-ATP, 4 Na-ATP, 0.4 Na-GTP, and 10 HEPES, pH 7.4 KOH. Bath saline contained the following (in mM): 140 NaCl, 2.5 KCl, 2.5 CaCl.sub.2, 2.5 MgCl.sub.2, 10 HEPES, and 11 glucose, pH 7.4 NaOH (Kontou et al., 2021, supra). All compound solutions were applied to cells using a three-barreled 700 μm pipe positioned just above the cell (Warner Instruments). Compound A was applied in lmg/mL 2-Hydroxypropyl-$-cyclodextrin, and its effects were compared to this vehicle alone. 10 mV voltage-ramp protocols over 1-s periods were used to determine the reversal potentials of the leak-subtracted muscimol-activated GABA.sub.AR currents. All voltages from whole-cell experiments were corrected offline using a calculated liquid junction potential value (13.6 mV) in Clampex (Molecular Devices). For gramicidin perforated-patch experiments, intracellular Cl.sup.− values were back-calculated using measured E.sub.GABA values and the Nernst equation:

[00001] $$E_{\text{ion}} = \frac{RT}{zF}\ln\frac{[\text{ion}]_o}{[\text{ion}]_i}$$

where E is the ion's reversal potential, R is the universal gas constant, T is temperature in kelvins, F is Faraday's constant, z is the charge of the ion, [chloride].sub.i is the intracellular chloride concentration and [chloride].sub.o is the extracellular ion concentration.

[0443] Measurements were performed in the presence of bumetanide (10 μM) and tetrodotoxin (TTX; 300 nM) to limit the contributions of NKCC1 and activity dependent shifts in chloride levels, respectively (Lee et al., 2011, supra). Cultures were exposed to Compound A or vehicle for 15 min and voltage ramps were used to determine changes in the reversal potential of GABA.sub.AR receptor currents over time (FIG. **3**A). These data were then used to determine shifts subsequently, using [Cl.sup.−] levels. At 15 min 300 nM Compound A (an approximately EC.sub.50 concentration) significantly reduced E.sub.GABA from −75±3.1 to −85.1±4.2 mV (FIG. **3**B; p=0.0104), a net −9.1±2.2 mV negative shift, reflecting a 2.2±0.6 mM reduction in Cl.sup.− (FIG. **3**C; p=0.0153). In contrast, vehicle did not significantly modify E.sub.GABA (FIG. **3**B; 0=−77.2±3.3, 15=−78.0±4.3 mV respectively, p=0.580) or chloride levels.

[0444] To measure the effects of Compound A on KCC2 activity under more dynamic conditions cultures in Compound A were incubated for 1 h and subjected to whole cell patch-clamp recording to artificially impose a 32 mM Cl.sup.− load on neurons via the patch pipette (Lee et al., 2022, supra; Moore et al., 2018, supra). 5 min after break-in, an initial E.sub.GABA value was determined and the effects of a subsequent 5 min incubation with VU0463271 was assessed. Compound A-treated cells exhibited lower basal E.sub.GABA values compared to controls (FIG. **3**C; FIG. **3**D −59.3±2.1 and −47.4±2.3 mv, respectively, p=0.025). Likewise, VU0463271 induced

shifts in E.sub.GABA were larger for cultures treated with Compound A (FIG. **3**C; FIG. **3**D 17.5±4.6 and 5.4±4.2 mV, respectively, p=0.009).
[0445] Thus, Compound A reduces neuronal Cl.sup.− levels and slows the development of hyperexcitability in an ex vivo system.
Example 5
Compound a Rapidly Accumulates in the Brain In Vivo
[0446] PK studies. Mice were injected intraperitoneally (IP), intravenously (IV), or subcutaneously (SC), with Compound A and 5% BCD in a maximum volume of 300 μl. Plasma and brain samples were rapidly frozen on dry ice. Tissue extracts were treated with Acetonitrile and Compound A levels were quantified using LC-MS/MS (Integrated Analytical Solutions, Inc, Berkeley, CA 94710; www.ianalytical.net).
[0447] To examine the ability of Compound A to cross the blood brain barrier, mice were injected with a single 25 mg/kg IV bolus; accumulation in the brain was evident within 30 minutes min in the brain and plasma to 7.6±4.4 and 41.5±11.5 and 11.2 μM respectively (FIG. **4**A). Likewise, a 50 mg/kg IP injection resulted in the accumulation of Compound A in the brain to 2.67±0.21 and 4.1±0.13 μM 2 and 4 hr later (FIG. **4**B). Drug accumulation in the brain was examined following SC injection (FIG. **4**C; 50 mg/kg). Accumulation was detected at 30 min and reached a maximal concentration of 675.75±80.5 nM at 4 hr, a level that was maintained 8 h later (FIG. **4**C).
[0448] Given the ability of Compound A to accumulate in the brain, it was examined to see if it also causes any gross effects on animal behavior. Mice were injected with Compound A (SC; 50 mg/kg), or vehicle and their behavior was compared in the open field test (Tretter et al., Proc. Natl. Acad. Sci. U.S.A 106, 20039-20044 (2009); Vien et al., Frontiers in molecular neuroscience 15, 817996 (2022)). Compound A did not modify the total distance traveled during the test trial (FIG. **4**D; Compound A=6134.2±235.3 cm, and vehicle=6453.7±215.4 cm; p=0.253, n=11 mice). Likewise, time spent in the center zone was comparable (FIG. **4**D; Compound A=18.4±2.1, and vehicle 16.6±2.5% of total time; p=0.658, n=11 mice).
[0449] Collectively, these results suggest that Compound A accumulates in the brain when dosed IV, IP or SC, and does not lead to any stereotyped effects on behavior.
[0450] It should be understood that the examples and embodiments provided herein are exemplary examples and embodiments. Those skilled in the art will envision various modifications of the examples and embodiments that are consistent with the scope of the disclosure herein. Such modifications are intended to be encompassed by the claims.

Claims

1. A method of treating neuropathic pain comprising administering a compound according to Formula (I), ##STR00037## or a pharmaceutically acceptable salt thereof, wherein: R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; A is selected from ##STR00038## or a N-oxide thereof; R.sup.3 is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl;

wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3 alkyl; R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6 alkyl; R.sup.10 is —C.sub.1-6 alkyl; R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3, to a subject diagnosed with the neuropathic pain in an amount of from about 0.01 mg to about 1500 mg.

2. The method of treating neuropathic pain according to claim 1, wherein the total amount of a compound according to Formula (I), or a pharmaceutically acceptable salt thereof, administered to the subject in a twenty-four hour period is between about 1 mg and about 1000 mg.

3. The method of treating neuropathic pain according to claim 1, wherein the neuropathic pain is chronic.

4. The method of treating neuropathic pain according to claim 1, wherein the neuropathic pain is acute.

5. The method of treating neuropathic pain according to claim 1, wherein the method provides improvement in at least one symptom selected from the group consisting of localized or generalized unpleasant bodily sensations, allodynia, hyperalgesia, burning, paresthesia and dysesthesia.

6. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound A: ##STR00039##

7. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound B: ##STR00040##

8. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound C: ##STR00041##

9. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound D: ##STR00042##

10. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound E: ##STR00043##

11. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound F: ##STR00044##

12. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound G: ##STR00045##

13. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is Compound H: ##STR00046##

14. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered enterally.

15. The method of treating neuropathic pain according to claim 14, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered orally, sublingually, buccally, transdermally or rectally.

16. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered parenterally.

17. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered in the form of a liquid for parenteral use, a tablet, a capsule, a caplet, a pill, an oral liquid, a lozenge, a film, a powder, an aerosol, or a patch.

18. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered as an extended release dosage form.

19. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered as an instant release dosage form.

20. The method of treating neuropathic pain according to claim 1, wherein the compound according to Formula (I), or a pharmaceutically acceptable salt thereof, is administered as a delayed release dosage form.

21. The method of treating neuropathic pain according to claim 1, wherein the subject has a spinal cord injury.

22. The method of treating neuropathic pain according to claim 1, wherein the subject has a peripheral nerve injury.

23. The method of treating neuropathic pain according to claim 1, wherein the subject has a traumatic brain injury.

24. The method of treating neuropathic pain according to claim 23, wherein the traumatic brain injury is a stroke.

25. The method of treating neuropathic pain according to claim 1, wherein the subject is human.

26. A method of treating neuro-inflammatory pain comprising administering a compound according to Formula (I), ##STR00047## or a pharmaceutically acceptable salt thereof, wherein: R.sup.1 is selected from C.sub.2-6alkyl; C.sub.2-6alkenyl; C.sub.2-6alkynyl; C.sub.2-6alkoxy; C.sub.2-6alkenyloxy; C.sub.2-6alkynyloxy; C.sub.3-7cycloalkyl; —O—C.sub.3-7cycloalkyl; C.sub.6-10aryl; —O—(CH.sub.2).sub.m—C.sub.6-10aryl; 6 membered heteroaryl; and thiophenyl; wherein alkyl, alkenyl, alkynyl, alkoxy, alkenyloxy, alkynyloxy and cycloalkyl are optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3 and wherein aryl and heteroaryl are optionally substituted with 1 or 2 substituents selected from -halo, —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy wherein —C.sub.1-3alkyl, —C.sub.1-8alkoxy and —C.sub.2-8alkynyloxy are optionally substituted with 1, 2, or 3 substituents selected from —F, —CF.sub.3, —NHC(O)O—C.sub.1-6alkyl or two substituents together with the carbon to which they are attached form diazirinyl; R.sup.2 is selected from —H; -halo; and —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; A is selected from ##STR00048## or a N-oxide thereof; R is selected from —H; —C.sub.1-6alkyl; —C.sub.2-6alkenyl; —C.sub.2-6alkynyl; C.sub.3-7cycloalkyl; and a 5 or 6 membered heterocycloalkyl; wherein the alkyl, alkenyl, alkynyl, cycloalkyl or heterocycloalkyl are optionally substituted by 1, 2 or 3 groups selected from —F, —CF.sub.3, —C.sub.1-3alkyl optionally substituted by 1 or 2 substituents selected from —F, —CF.sub.3, —C(O)NR.sup.8R.sup.9 and —NR.sup.8R.sup.9; R.sup.4a and R.sup.4b are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.4c and R.sup.4d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3, or R.sup.4c and R.sup.4d together with the carbon to which they are attached represent carbonyl; R.sup.5a, R.sup.5b, R.sup.5c and R.sup.5d are each independently selected from —H and —C.sub.1-3 alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; R.sup.6 is selected from —H; -halo; —NH.sub.2; —CN; —C.sub.1-3alkyl optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and —CF.sub.3; —C(O)O—C.sub.1-3alkyl; —C(O)NR.sup.8R.sup.9; —C(O)OH; and —NHC(O)—C.sub.1-3 alkyl; R.sup.7 is selected from NR.sup.10R.sup.11; a 5 to 7 membered monocyclic heterocycloalkyl; and a 5 or 6 membered monocyclic heteroaryl; wherein the heterocycloalkyl and heteroaryl are optionally substituted with 1, 2 or 3 groups selected from —CN; —C.sub.1-6alkyl optionally substituted with 1, 2 or 3 substituents selected from —F, —CF.sub.3 and —OH; —C.sub.1-3alkoxy optionally substituted with 1, 2 or 3 substituents selected from —F and CF.sub.3; —C(O)OH; —C.sub.1-3alkylene-NHC(O)C.sub.1-6alkyl; —C.sub.1-3alkylene-NHC(O)OC.sub.1-6alkyl; C.sub.3-5cycloalkyl; or the heterocycloalkyl is optionally substituted with two substituents on the same ring carbon which together with the carbon atom to which they are attached form a 5 to 7 membered monocyclic heterocycloalkyl; and wherein when R.sup.7 is morpholinyl and R.sup.1 is unsubstituted phenyl, R.sup.2 is not —H; R.sup.8 and R.sup.9 are each independently selected from —H and —C.sub.1-6 alkyl; R.sup.10 is —C.sub.1-6 alkyl; R.sup.11 is selected from —C.sub.1-6alkyl optionally substituted with 1 or 2 substituents selected from —F and —C.sub.1-3alkoxy; and —(CH.sub.2).sub.nR.sup.12; R.sup.12 is a 5 or 6 membered heteroaryl, a 3 to 5 membered cycloalkyl or a 3 to 6 membered heterocycloalkyl; m is 0 or 1; and n is 1, 2 or 3, to a subject diagnosed with the neuro-inflammatory pain in an amount of from about 0.01 mg to about 1500 mg.